First Midterm Exam

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3
Ander Aguirre (MWF 9:55 AM)	Sarah Tammen (TR 9:30 AM)	Nicholas Derr (MWF 9:50 AM)

- **There are 5 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
Total:	60	

1. Suppose we perform an experiment in which we sample 3 balls without replacement from an urn filled with red, white, and green balls. Suppose that there are twenty balls total, and suppose for convenience that they are numbered; specifically, suppose that the red balls are numbered 1-10, that the white balls are numbered 11-15, and that the green balls are numbered 16-20. Suppose we write down the numbers on the balls in the order we draw them.

   (a) (4 points) What is the sample space of this experiment? If we let our set of events consist of all outcomes of the sample space, what is an appropriate probability measure?

   Solution:
   Since we are sampling with order, our sample space should consist of ordered triples of integers. Since we sample without replacement, all three coordinates of each triple should be distinct. Thus, we take

   $$\Omega = \{(x_1, x_2, x_3) : x_1, x_2, x_3 \in \{1, 2, \ldots, 20\} \text{ with } x_i \neq x_j \text{ if } i \neq j\}.$$

   Each triple has an equal probability of being chosen. Thus for any triple (x_1, x_2, x_3), we have

   $$P((x_1, x_2, x_3)) = \frac{1}{\#\Omega} = \frac{1}{20 \cdot 19 \cdot 18},$$

   and any event A has probability

   $$P(A) = \frac{\#A}{\#\Omega} = \frac{\#A}{20 \cdot 19 \cdot 18}.$$

   (b) (6 points) Determine the probability of each of the following events. You may leave your answers unsimplified.

      i. The event that all three balls in the sample are red

      Solution:
      The probability of interest is

      $$\frac{10 \cdot 9 \cdot 8}{20 \cdot 19 \cdot 18}.$$

      One way to see this is as follows: if we let R_i be the event that ball i is red for $i = 1, 2, 3$, then we have that

      $$P(R_1 R_2 R_3) = P(R_1)P(R_2|R_1)P(R_3|R_1 R_2) = \frac{10}{20}\left(\frac{9}{19}\right)\left(\frac{8}{18}\right).$$

      Note, however, that we gave full credit for a correct answer with no justification. Note also that the above expression is equivalent to

      $$\frac{\binom{10}{3}}{\binom{20}{3}},$$

      so we gave full credit for this as well.

      ii. The event that none of the balls in the sample are red

      Solution:
      The probability of interest is again

      $$\frac{10 \cdot 9 \cdot 8}{20 \cdot 19 \cdot 18}.$$

If we borrow the notation from part (i), then we can see this as follows:

$$P(R_1^c R_2^c R_3^c) = P(R_1^c)P(R_2^c|R_1^c)P(R_3^c|R_1^c R_2^c) = \frac{10}{20}\left(\frac{9}{19}\right)\left(\frac{8}{18}\right).$$

iii. The event that the sample contains balls of at least two different colors

Solution:
We have that

$$\begin{aligned} P(\{\text{at least two different colors}\}) &= 1 - P(\{\text{all balls are the same color}\}) \\ &= 1 - (P(\{\text{all balls red}\}) - P(\{\text{all balls green}\}) - P(\{\text{all balls white}\})) \\ &= 1 - \frac{10 \cdot 9 \cdot 8}{20 \cdot 19 \cdot 18} - 2\left(\frac{5 \cdot 4 \cdot 3}{20 \cdot 19 \cdot 18}\right). \end{aligned}$$

2. Suppose we are given a pair of fair tetrahedral dice. (A tetrahedral die is a triangular pyramid with the faces numbered 1-4.) Let X_1 be the first roll, let X_2 be the second roll, and let S be the sum of the rolls.

(a) (3 points) Determine the probability mass function of S.

Solution:
The possible values of S are $2, 3, 4, 5, 6, 7, 8$, and S has probability mass function

$$p_S(k) = \begin{cases} \frac{1}{16}, & k \in \{2, 8\}; \\ \frac{2}{16} = \frac{1}{8}, & k \in \{3, 7\}; \\ \frac{3}{16}, & k \in \{4, 6\}; \\ \frac{4}{16}, & k = 5. \end{cases}$$

One way to see this is to make a chart whose (i, j)-entry is $P(X_1 = i, X_2 = j)$ for $i, j \in \{1, 2, 3, 4\}$.

(b) (7 points) (2+5) Suppose we roll the dice repeatedly until the sum of the rolls is 4 or 6 and agree that I will win if we see a sum of 4 first but you will win if we see a sum of 6 first.

i. Let N be the number of rounds required to determine a winner, where a round consists of rolling both dice. What is the distribution of N? Identify the name of the distribution and all relevant parameters.

Solution:
We have that $N \sim \text{Geom}\left(\frac{3}{8}\right)$ so that N has probability mass function

$$p_N(k) = \left(\frac{5}{8}\right)^{k-1}\left(\frac{3}{8}\right), \qquad k \in \mathbb{N}.$$

ii. Determine, with justification, the probability that the probability that the game ends on or before the third round with a sum of six.
Solution:
Let A be the event that the game ends with a sum of 6. For each $i \in \mathbb{N}$, let B_i be the event that $N = i$. We are interested in computing $P(A \cap \{N \leq 3\})$. Since the event that $N \leq 3$ is the disjoint union $B_1 \cup B_2 \cup B_3$), we have that

$$\begin{aligned} P(A \cap \{N \leq 3\}) &= P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) \\ &= P(A \cap B_1) + P(A|B_1^c)P(B_1^c) + P(A|B_1^c B_2^c)P(B_1^c B_2^c) \\ &= \frac{3}{16} + \frac{3}{16}\left(\frac{5}{8}\right) + \frac{3}{16}\left(\frac{5}{8}\right)^2 \\ & \frac{3}{16}\left(1 + \frac{5}{8} + \left(\frac{5}{8}\right)^2\right). \end{aligned}$$

3. Throughout this question, let A, B, C be three *mutually independent* events in a probability space with $P(A) = P(B) = P(C) = p$.

(a) (3 points) Show that $A \cap C$ is independent of B^c

Solution:

By mutual independence:

$$P((A \cap B^c) \cap C) = P(A \cap B^c \cap C) = P(AC) - P(ABC) = p^2 - p^3$$

Also by mutual independence:

$$P(A \cap B^c)P(C) = [P(A) - P(AB)]P(C) = p^2 - p^3$$

(b) (7 points) Compute, in terms of p, the following conditional probability

$$P(A \cap B^c \cap C \mid A \cup B^c \cup C).$$

Solution:

By mutual independence, Bayes theorem and inclusion-exclusion:

$$\begin{aligned} P(A \cap B^c \cap C \mid A \cup B^c \cup C) &= \frac{P(A)P(B^c)P(C)}{P(A) + P(B) + P(C) - P(AB^c) - P(B^cC) - P(AC) + P(AB^cC)} \\ &= \frac{p^2 - p^3}{2p + (1-p) - 2p(1-p) - p^2 + p^2(1-p)} \end{aligned}$$

4. We follow a state election in two counties: Adams and Brown. Our candidate either wins the state (W) or loses the state (L). Before any county results are reported we estimate even odds:

$$P(W) = 0.5, \qquad P(L) = 0.5.$$

Denote the events that our candidate wins Adams by A and Brown by B. Historical data shows:

$$P(A \mid W) = 0.8, \qquad P(B \mid W) = 0.6,$$

$$P(A \mid L) = 0.3, \qquad P(B \mid L) = 0.4.$$

Assume county outcomes are *conditionally independent* given W or L.

(a) (4 points) Suppose Adams reports first and our candidate wins it. Compute $P(W \mid A)$.

Solution:

$$P(W|A) = \frac{P(A|W)P(W)}{P(A|W)P(W) + P(A|L)P(L)} = \frac{0.4}{0.4 + 0.15} = \frac{8}{11}$$

(b) (6 points) Suppose our candidate wins Adams but loses Brown. Compute $P(W \mid A \cap B^c)$.

Solution:

Once again using Bayes theorem as well as conditional independence:

$$\begin{aligned} P(W|AB^C) &= \frac{P(A|W)P(B^c|W)P(W)}{P(A|W)P(B^c|W)P(W) + P(A|L)P(B^c|L)P(L)} \\ &= \frac{(0.8)(0.4)(0.5)}{(0.8)(0.4)(0.5) + (0.3)(0.6)(0.5)} = \frac{16}{25} \end{aligned}$$

5. Consider a random variable Y with probability density function

$$P(Y = y) = \begin{cases} y, & y \in [\frac{1}{2}, a), \\ 0, & \text{else.} \end{cases}$$

(a) (2 points) Find a such that $p_Y(y)$ is a valid probability distribution function.

Solution:

We require that the probability density function $f_Y(y)$ integrates to 1 over all possible values, so

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} f_Y(y)dy = \int_{1/2}^{a} ydy \\ &= \frac{y^2}{2}\Big]_{1/2}^{a} = \frac{a^2}{2} - \frac{1}{8} \\ &\implies \frac{9}{8} = \frac{a^2}{2} \\ &\implies \frac{9}{4} = a^2 \\ &\implies \boxed{a = \frac{3}{2}.} \end{aligned}$$

(b) (3 points) Introduce a new discrete random variable $U = \lfloor 2Y \rfloor$, where the floor operator $z = \lfloor r \rfloor$ represents the largest integer $z \leq r$ for $r \in \mathbb{R}$. Calculate the probability mass distribution of U.

Solution:

For each possible value u of U, the probability $P(U = u)$ will satisfy

$$P(U = u) = \int_{\{y : \lfloor 2y \rfloor = u\}} f_Y(y)dy,$$

i.e. we need to determine which y-values will yield that value of U and integrate the probability density over that range to obtain a probability mass. In this case, $y \in [\frac{1}{2}, 1)$ will correspond to $U = 1$, and $y \in [1, \frac{3}{2})$ to $U = 2$. The probability of U taking any other value is 0. Thus,

$$P(U = 1) = \int_{1/2}^{1} ydy = \frac{1^2}{2} - \frac{\left(\frac{1}{2}\right)^2}{2} = \frac{1}{2} - \frac{1}{8} \implies \boxed{P(U = 1) = \frac{3}{8},}$$

$$P(U = 2) = \int_{1}^{3/2} ydy = \frac{\left(\frac{3}{2}\right)^2}{2} - \frac{1^2}{2} = \frac{9}{8} - \frac{4}{8} \implies \boxed{P(U = 2) = \frac{5}{8}.}$$

(c) (2 points) Find b such that $P(Y < b) = 1/2$.

Solution:

For the probability density function $f_Y(y)$, we must have

$$P(Y \leq b) = \int_{-\infty}^{b} f_Y(y)dy = \int_{1/2}^{b} y\ dy = \frac{b^2}{2} - \frac{1}{8}.$$

Note that we know $b < \frac{3}{2}$, allowing us to just use y (not the full multiple-case function) as the integrand. Setting this value equal to $\frac{1}{2}$ as requested,

$$\frac{1}{2} = \frac{b^2}{2} - \frac{1}{8} \implies \frac{5}{8} = \frac{b^2}{2} \implies \frac{5}{4} = b^2 \implies \boxed{b = \frac{\sqrt{5}}{2}.}$$

(d) (3 points) Consider the events $A = \{Y \in [5/8, 12/8]\}$ and $B = \{Y \in [1, 2]\}$. Are they independent?

Solution:

We can compute the probability of each event as well as their intersection:

$$P(A) = \int_{5/8}^{12/8} f_Y(y)dy = \int_{5/8}^{3/2} y\ dy = \frac{\left(\frac{3}{2}\right)^2}{2} - \frac{\left(\frac{5}{8}\right)^2}{2} = \frac{9}{8} - \frac{25}{128} = \frac{144}{128} - \frac{25}{128} = \frac{119}{128} < 1.$$

We could also just have noted that $P(A) < 1$, since the interval corresponding to A doesn't include all possible values of Y. We also have

$$P(B) = \int_{1}^{2} f_Y(y) = \int_{1}^{3/2} y\ dy = \frac{5}{8}.$$

Finally, note that $B \implies A$, since there is no probability of $Y \in [\frac{3}{2}, 2]$, and the rest of B's interval lies within that of A. Thus,

$$P(AB) = P(B) = \frac{5}{8},$$

and since $P(A) < 1$, we have $\boxed{P(AB) = P(B) \neq P(A)P(B),}$ indicating they are **not** independent.

6. Consider the following experiment: you roll a six-sided die until you obtain a prime number (2, 3, or 5). Then, you perform a number of coin flips equal to the number of times you rolled the die. We define the random variables N as the number of rolls/flips and T as the number of tails obtained.

(a) (2 points) For each $n \in \mathbb{N}$, let B_n be the event that $N = n$. For a given n, what is the conditional probability $P(T \geq 1|B_n)$?

Solution:

First, we note that the complement may be easier to calculate, since

$$P(T = 0|B_n) = 1 - P(T \geq 1|B_n).$$

If the event B_n occurs, then $N = n$ and $(T|B_n) \sim \text{Bin}\left(n, \frac{1}{2}\right)$, so

$$P(T = 0|B_n) = \binom{n}{0}\left(\frac{1}{2}\right)^n = 2^{-n}.$$

Thus,

$$\boxed{P(T \geq 1|B_n) = 1 - 2^{-n}.}$$

(b) (3 points) What is the unconditional probability $P(T \geq 1)$?

Solution:

To obtain the unconditional probability, we must sum over all possible n, i.e.

$$P(T \geq 1) = \sum_{n=1}^{\infty} P(T \geq 1|B_n)P(B_n).$$

We know that $B_n \sim \text{Geom}\left(\frac{1}{2}\right)$, since there is probability 1/2 of rolling a 2, 3 or 5. Thus, $P(B_n) = 2^{-n}$ ($n - 1$ factors of 1/2 for $n - 1$ heads and one more for the final tails), and

$$P(T \geq 1) = \sum_{n=1}^{\infty}(1 - 2^{-n})2^{-n} = \sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n - \sum_{n=1}^{\infty}\left(\frac{1}{4}\right)^n.$$

We can evaluate the series as

$$\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n = \frac{1}{2}\sum_{n=0}\left(\frac{1}{2}\right)^n = \frac{1}{2}\frac{1}{1-\frac{1}{2}} = 1,$$

$$\sum_{n=1}^{\infty}\left(\frac{1}{4}\right)^n = \frac{1}{4}\sum_{n=0}\left(\frac{1}{4}\right)^n = \frac{1}{4}\frac{1}{1-\frac{1}{4}} = \frac{1}{3},$$

so we conclude

$$\boxed{P(T \geq 1) = 1 - \frac{1}{3} = \frac{2}{3}.}$$

(c) (5 points) Give all possible values k of T and find the probability mass function $P(T = k)$. It may be useful to note that for $0 \leq p < 1$,

$$\sum_{k=0}^{\infty} \binom{k}{i} p^k = \frac{p^i}{(1-p)^{i+1}},$$

using the convention that $\binom{k}{i} = 0$ for $i < 0$ and $i > k$.

Solution:

We expect $T \in \mathbb{N} \cup \{\infty\}$, albeit with $P(T = \infty) = 0$. For $T = k$ we will need to sum over all possible n, i.e.

$$P(T = k) = P(\{T = k\} \cup B_n) = P(T = k|B_n)P(B_n).$$

This corresponds to thinking about all of the possible N with which we could have obtained that k. We already know $P(B_n) = 2^{-n}$. Since we know $T|B_n$ is binomial with n trials with $p = \frac{1}{2}$, we have

$$P(T = k|B_n) = \binom{n}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{n-k} = \binom{n}{k} \left(\frac{1}{2}\right)^n.$$

Thus we have

$$P(T = k) = \sum_{n=1}^{\infty} \binom{n}{k} \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n.$$

Note that since $\binom{n}{k} = 0$ for $k > n$, the terms in the sum will vanish for $n = 0, 1, 2, \ldots, k-1$. Specifically, for $k > 0$, the $n = 0$ term vanishes and the sum from $n = 1$ is equal to the sum from $n = 0$ (corresponding to the provided expression.) For the $k = 0$ term, we must correct for the difference, and we write

$$P(T = k) = \begin{cases} \sum_{n=0}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n, & k \geq 1, \\ -1 + \sum_{n=0}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n, & k = 0. \end{cases}$$

Now, we note that when we sub in the provided relation (with $k = n$, $i = k$, $p = 1/4$) we get

$$\sum_{n=0}^{\infty} \binom{n}{k} \left(\frac{1}{4}\right)^n = \frac{\left(\frac{1}{4}\right)^k}{\left(1 - \frac{1}{4}\right)^{1+k}} = \frac{4}{3^{1+k}}.$$

This, then, is our answer for $P(T = k)$ with $k > 0$, but for $k = 0$ we correct to get

$$P(T = 0) = -1 + \frac{4}{3} = \frac{1}{3}.$$

Thus, ultimately we get

$$\boxed{P(T = k) = \begin{cases} 4/3^{1+k}, & k \geq 1, \\ 1/3, & k = 0. \end{cases}}$$

So the first few values are

k	0	1	2	3	$\cdots$
$P(T = k)$	$\frac{1}{3}$	$\frac{4}{9}$	$\frac{4}{27}$	$\frac{4}{81}$	$\cdots$

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List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

First Midterm Exam

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
D. Kwon (TuTh 1PM)	Benedek Valkó (MWF 11AM)	T. Shcherbyna (TuTh 11AM)	D. Clancy (MWF 2:25PM)

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- Please present your solutions in a clear manner. **Justify your steps.** (Unless it is explicitly stated that this is not required.) In problems 2.-5. numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
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Problem	Points
1	/20
2	/15
3	/18
4	/23
5	/24
Total	/100

1. **Short problems. (4 points each)** In the following problems (but only here!) you will not need to justify your answer.

   (a) We sample three numbers without replacement with order from the set $\{1, 2, \ldots, 100\}$. Describe the sample space of the experiment.

   (b) Suppose that X has geometric distribution with parameter $p = 1/3$. What is the probability that X is equal to 4 or 5?

   (c) We shuffle a pack of 52 cards, and choose 5 cards (without replacement). Let Z be the number of aces among the 5 cards. Identify the distribution of Z.
   (Either provide a name and parameters, or the probability mass function.)

(Problem 1 continued)

(d) The random variable Z has probability density function given by

$$f(z) = \begin{cases} \frac{2}{\pi}\sqrt{1-z^2}, & \text{if } -1 \leq z \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Compute the probability $P(Z < 5)$.

(e) Let U be a Unif$[0, 2]$ distributed random variable. Decide whether the events $A = \{U \in [0, 1]\}$ and $B = \{U \in [2/3, 4/3]\}$ are independent or not.

2. Three events A, B and C satisfy the following: any two of them are independent, the intersection of the three events is the empty set, and we have $P(A) = P(B) = P(C) = 1/4$.

   (a) **(5 points)** Give the precise definition for the mutual independence of three events, and decide whether A, B, and C are mutually independent or not.

   (b) **(10 points)** Find $P(A \cup B \cup C)$.

3. Brianna is trick-or-treating for Halloween, and she arrives to a house which has two baskets full of treats at the front door. The first basket has 20 chocolate bars and 30 hard candies. The second basket has 15 chocolate bars and 45 hard candies. Brianna chooses one of the two baskets randomly, and then she chooses a treat from the chosen basket (also randomly).

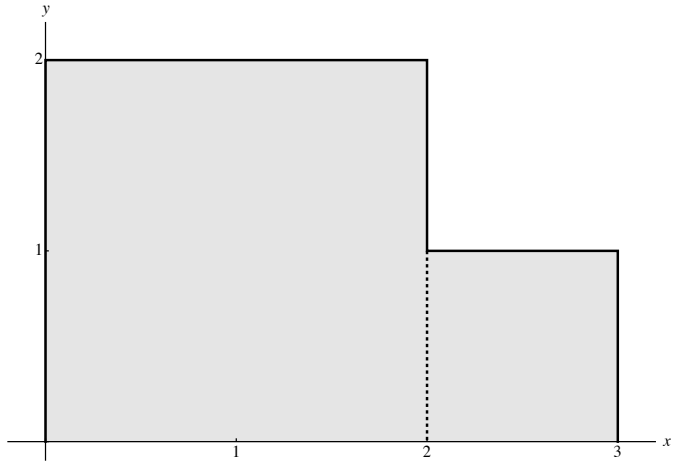
   (a) **(5 points)** What is the probability that Brianna picks a hard candy?

   (b) **(8 points)** We learn that Brianna picked a hard candy. What is the probability that she chose it from the first basket?

(Problem 3 continued)

(c) **(5 points)** Brianna picked a hard candy, but she doesn't like the flavor. She decides to put it back in the basket (the same one she picked it from), and she chooses a candy randomly again from the same basket. It turns out that she got a hard candy for her second pick again (maybe the same one, maybe not). What is the probability that Brianna picked from the first basket?
(You do not need to simplify your answer here.)

4. The shaded region A in the picture below is made from the union of a square of side length 2 and a square of side length 1.



We choose a point uniformly at random from the region A. Let X be the x-coordinate of the chosen point.

(a) **(4 points)** What is the range of X?

(b) **(4 points)** Find the value of $P(X < 5/2)$.

(Problem 4 continued)

(c) **(8 points)** Find the CDF of X.
Hint: it will be a case-defined function!

(d) **(7 points)** Explain why X is continuous random variable, and find its PDF.

5. You have two dice in front of you: a regular 6-sided fair die and a 4-sided fair die (with faces numbered with 1,2,3,4). You repeatedly roll both dice simultaneously.

   (a) **(5 points)** Let p be the the probability that you roll at least one 4 on the first roll of the two dice. Find the value of p.

   (b) **(5 points)** Let X be the number of times you see at least one 4 on your first 10 rolls of the two dice. What is the distribution of X?

(Problem 5 continued)

(c) **(5 points)** You keep on rolling the pair of dice until you roll at least one 4. Let N be the number of times you roll the dice. ($N = 1$ means that you rolled at least one 4 on the first roll of the two dice.) What is the probability that N is larger than 5?
Express your answer in terms of the probability p defined in part (a).

(d) **(5 points)** What is the probability that on the Nth roll both dice show four?
(The random variable N is defined in part (c).)

(Problem 5 continued)

(e) **(4 points)** Let N be defined as in part (c), and define the random variable Y as follows:

$$Y = \begin{cases} 1 & \text{if on the } N\text{th roll only the 6-sided die shows 4,} \\ 2 & \text{if on the } N\text{th roll only the 4-sided die shows 4,} \\ 3 & \text{if on the } N\text{th roll both dice show 4} \end{cases}$$

Decide whether N and Y are independent or not.

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First Midterm Exam – Solutions

1. **Short problems. (4 points each)** In the following problems (but only here!) you will not need to justify your answer.

   (a) We sample three numbers without replacement with order from the set $\{1, 2, \ldots, 100\}$. Describe the sample space of the experiment.

   Solution:
   $$\Omega = \{(a_1, a_2, a_3) : 1 \le a_i \le 100, 1 \le i \le 3, \text{ and } a_i \ne a_j \text{ for } i \ne j\}$$

   (b) Suppose that X has geometric distribution with parameter $p = 1/3$. What is the probability that X is equal to 4 or 5?

   Solution:
   $$\begin{aligned} P(X \text{ is 4 or 5}) &= P(X = 4) + P(X = 5) = p(1-p)^3 + p(1-p)^4 = (1/3)(2/3)^3 + (1/3)(2/3)^4 \\ &= \frac{40}{243} \end{aligned}$$

   (c) We shuffle a pack of 52 cards, and choose 5 cards (without replacement). Let Z be the number of aces among the 5 cards. Identify the distribution of Z.
   (Either provide a name and parameters, or the probability mass function.)

   Solution: Z has hypergeometric distribution with $N = 52$, $N_A = 4$, $n = 5$. The probability mass function is
   $$P(Z = k) = \frac{\binom{4}{k}\binom{48}{5-k}}{\binom{52}{5}}$$
   for $k = 0, 1, 2, 3, 4$.

   (d) The random variable Z has probability density function given by
   $$f(z) = \begin{cases} \frac{2}{\pi}\sqrt{1 - z^2}, & \text{if } -1 \le z \le 1 \\ 0, & \text{otherwise.} \end{cases}$$
   Compute the probability $P(Z < 5)$.

   Solution: $P(Z < 5) = 1 - P(Z > 5) = 1 - \int_5^\infty f(z)dz = 1$. (Since f is 0 for $z > 5$.)
   OR: The PDF is zero outside of the interval $[-1, 1]$, hence $-1 \le Z \le 1$ with probability one. Since $P(Z < 5) \ge P(Z \in [-1, 1]) = 1$, this implies $P(Z < 5) = 1$.
   OR: $P(Z < 5) = \int_{-\infty}^5 f(z)dz = \int_{-1}^1 f(z)dz = \int_{-\infty}^\infty f(z)dz = 1$ (since f is zero outside of $[-1, 1]$.)

   (e) Let U be a Unif$[0, 2]$ distributed random variable. Decide whether the events $A = \{U \in [0, 1]\}$ and $B = \{U \in [2/3, 4/3]\}$ are independent or not.

   Solution: We have $AB = \{U \in [2/3, 1]\}$, and
   $$P(A) = \frac{1}{2}, \qquad P(B) = \frac{2/3}{2} = \frac{1}{3}, \qquad P(AB) = \frac{1/3}{2} = \frac{1}{6}.$$
   Since $P(AB) = P(A)P(B)$, we have that A and B are independent.

2. Three events A, B and C satisfy the following: any two of them are independent, the intersection of the three events is the empty set, and we have $P(A) = P(B) = P(C) = 1/4$.

   (a) **(5 points)** Give the precise definition for the mutual independence of three events, and decide whether A, B, and C are mutually independent or not.

   Solution: The events A, B and C are independent if all of the following equations hold:

   $$P(AB) = P(A)P(B), \quad P(AC) = P(A)P(C), \quad P(BC) = P(B)P(C), \quad P(ABC) = P(A)P(B)P(C).$$

   In our case $ABC = \emptyset$, hence $P(ABC) = 0 \neq P(A)P(B)P(C) = \frac{1}{4^4}$, which shows that these three events are not mutually independent.

   (b) **(10 points)** Find $P(A \cup B \cup C)$.

   Solution: Using the inclusion-exclusion formula for three events:

   $$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC).$$

   By the pairwise independence of the events we know that

   $$P(AB) = P(A)P(B) = \frac{1}{4^2}, \quad P(AC) = P(A)P(C) = \frac{1}{4^2}, \quad P(BC) = P(B)P(C) = \frac{1}{4^2}.$$

   We also have $P(ABC) = P(\emptyset) = 0$. This gives

   $$P(A \cup B \cup C) = 3 \cdot \frac{1}{4} - 3 \cdot \frac{1}{4^2} = \frac{9}{16}.$$

3. Brianna is trick-or-treating for Halloween, and she arrives to a house which has two baskets full of treats at the front door. The first basket has 20 chocolate bars and 30 hard candies. The second basket has 15 chocolate bars and 45 hard candies. Brianna chooses one of the two baskets randomly, and then she chooses a treat from the chosen basket (also randomly).

(a) **(5 points)** What is the probability that Brianna picks a hard candy?

Solution: Let A be the event that Brianna chose from the first basket, and B the event that she chose a hard candy. By the setup of the problem we know that

$$P(A) = P(A^c) = 1/2, \quad P(B|A) = \frac{30}{50} = \frac{3}{5}, \quad P(B|A^c) = \frac{45}{60} = \frac{3}{4}.$$

Our goal is to compute $P(B)$.

Using $P(B) = P(BA) + P(BA^c)$ and the properties of conditional probability we get

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c) = \frac{3}{5} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{12+15}{40} = \frac{27}{40}.$$

(b) **(8 points)** We learn that Brianna picked a hard candy. What is the probability that she chose it from the first basket?

Solution: Using Bayes' formula

$$\begin{aligned} P(A|B) &= \frac{P(AB)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)} \\ &= \frac{\frac{3}{5} \cdot \frac{1}{2}}{\frac{3}{5} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2}} = \frac{\frac{3}{10}}{\frac{27}{40}} = \frac{4}{9}. \end{aligned}$$

(c) **(5 points)** Brianna picked a hard candy, but she doesn't like the flavor. She decides to put it back in the basket (the same one she picked it from), and she chooses a candy randomly again from the same basket. It turns out that she got a hard candy for her second pick again (maybe the same one, maybe not). What is the probability that Brianna picked from the first basket?
(You do not need to simplify your answer here.)

Solution: Let C be the event that Brianna picked a hard candy for the second pick. We may assume that B and C are *conditionally* independent of A (and A^c): if we already know which basket Brianna is picking from then her first and second pick are independent.

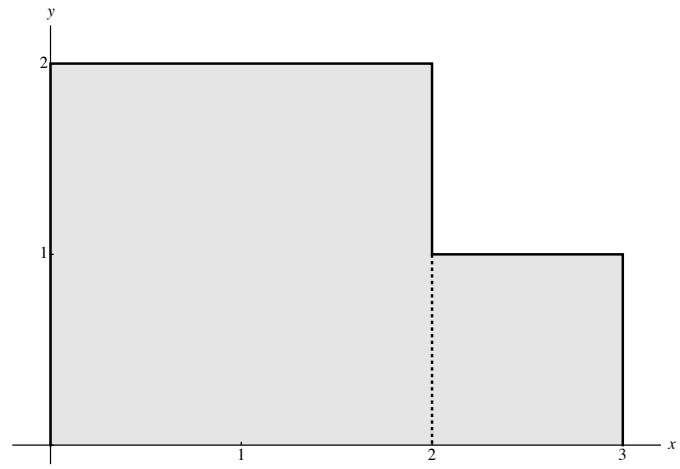
Hence

$$P(BC|A) = P(B|A)P(C|A) = \left(\frac{3}{5}\right)^2, \qquad P(BC|A^c) = P(B|A^c)P(C|A^c) = \left(\frac{3}{4}\right)^2$$

We need to compute $P(A|BC)$, we can do that as follows:

$$P(A|BC) = \frac{P(ABC)}{P(BC)} = \frac{P(BC|A)P(A)}{P(BC|A)P(A) + P(BC|A^c)P(A^c)} = \frac{\frac{1}{2}\left(\frac{3}{5}\right)^2}{\frac{1}{2}\left(\frac{3}{5}\right)^2 + \frac{1}{2}\left(\frac{3}{4}\right)^2} = \frac{16}{41}$$

4. The shaded region A in the picture below is made from the union of a square of side length 2 and a square of side length 1.



We choose a point uniformly at random from the region A. Let X be the x-coordinate of the chosen point.

(a) **(4 points)** What is the range of X?

Solution: If the point is in the large square then the x-coordinate of the point is in $[0, 2]$, if it is in the small square then it is in $[2, 3]$. Hence the range of X is $[0, 3]$ (and all of these values are possible).

(b) **(4 points)** Find the value of $P(X < 5/2)$.

Solution: The area of the sample space is $2^2 + 1^2 = 5$. The probability that the chosen point is in some sub-region B of A is proportional to the area of B: $\frac{\text{area}(B)}{5}$.

The event $\{X < 5/2\}$ is the same as the point being in the shaded region (which is the union of the large square and a rectangle with sides 1 and 1/2). The area of this region is $4 + 1/2$, hence

$$P(X < 5/2) = \frac{4 + 1/2}{5} = \frac{9}{10}.$$

(c) **(8 points)** Find the CDF of X.

Hint: it will be a case-defined function!

Solution: We need to compute $F(x) = P(X \leq x)$ for all x.

Since the range of X is $[0, 3]$, we have $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x \geq 3$.

If $0 \leq x \leq 2$ then the region corresponding to $\{X \leq x\}$ is a rectangle with sides x and 2, hence in this case $F(x) = P(X \leq x) = \frac{2 \cdot x}{5}$.

If $2 < x \leq 3$ then the region corresponding to $\{X \leq x\}$ is the union of the large square and a rectangle with sides $x - 2$ and 1, hence in this case $F(x) = P(X \leq x) = \frac{4+(x-2)}{5} = \frac{2+x}{5}$.

The CDF is given by the following case-defined function:

$$F(x) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{2x}{5}, & \text{if } 0 \leq x \leq 2 \\ \frac{2+x}{5}, & \text{if } 2 < x \leq 3 \\ 1, & \text{if } 3 < x. \end{cases}$$

(d) **(7 points)** Explain why X is continuous random variable, and find its PDF.

Solution: The CDF found in part (d) is a continuous function (we only need to check this for $x = 0, 2$ and 3), and it's differentiable everywhere except at 0, 2, and 3. Hence X is a continuous random variable and we can find its PDF by differentiating. (At $x = 0, 2, 3$ we can choose the value as we wish.) Hence the PDF is

$$f(x) = \begin{cases} \frac{2}{5}, & \text{if } 0 < x < 2 \\ \frac{1}{5}, & \text{if } 2 < x < 3 \\ 0, & \text{otherwise.} \end{cases}$$

You can also find the PDF directly by the infinitesimal method. We only need to worry about values between 0 and 3, since outside the range the PDF is 0. If $0 < a < 2$ and ε small then $P(X \in [a, a+\varepsilon]) = \frac{2\varepsilon}{5}$, since the region corresponding to $\{X \in [a, a+\varepsilon]\}$ is a rectangle with sides ε and 2. If $2 < a < 3$ then $P(X \in [a, a+\varepsilon]) = \frac{1\varepsilon}{5}$, since the region corresponding to $\{X \in [a, a+\varepsilon]\}$ is a rectangle with sides ε and 1. This shows that the PDF is 2/5 in $(0, 2)$ and it is 1/5 in $(2, 3)$, giving the same answer we found before.

5. You have two dice in front of you: a regular 6-sided fair die and a 4-sided fair die (with faces numbered with 1,2,3,4). You repeatedly roll both dice simultaneously.

(a) **(5 points)** Let p be the the probability that you roll at least one 4 on the first roll of the two dice. Find the value of p.

Solution: Let $A = \{\text{first roll of the regular die is } 4\}$, $B = \{\text{first roll of the four-sided die is } 4\}$. These events are independent, and $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{4}$. We need to compute $P(A \cup B)$. We can do that using inclusion-exclusion (and independence):

$$P(A \cup B) = P(A) + P(B) - P(AB) = \frac{1}{6} + \frac{1}{4} - \frac{1}{6} \cdot \frac{1}{4} = \frac{3}{8}$$

or by considering the complement, and using the independence of A^c, B^c:

$$P(A \cup B) = 1 - P(A^c B^c) = 1 - P(A^c)P(B^c) = 1 - \frac{5}{6} \cdot \frac{3}{4} = \frac{3}{8}.$$

(b) **(5 points)** Let X be the number of times you see at least one 4 on your first 10 rolls of the two dice. What is the distribution of X?

Solution: We can consider the rolls as trials, with success being getting at least one 4 from the pair of dice. Then X counts the number of successes among 10 repeated independent trials with success probability $p = 3/8$, which means that X has binomial distribution with parameters $n = 10$, $p = 3/8$.

(c) **(5 points)** You keep on rolling the pair of dice until you roll at least one 4. Let N be the number of times you roll the dice. ($N = 1$ means that you rolled at least one 4 on the first roll of the two dice.) What is the probability that N is larger than 5?
Express your answer in terms of the probability p defined in part (a).

Solution: N is a geometric random variable with success probability $p = 3/8$. We can compute $P(N > 5)$ by breaking the event $\{N > 5\}$ up according to the possible values of N:

$$P(N > 5) = \sum_{k=6}^{\infty} P(N = k) = \sum_{k=6}^{\infty} p(1-p)^{k-1} = p(1-p)^5 \sum_{j=0}^{\infty} (1-p)^j = \frac{p(1-p)^5}{1-(1-p)} = (1-p)^5.$$

We can get the same result a bit quicker by noting that $\{N > 5\}$ is the same event that the first 5 trials were failures (i.e. there were no fours among the first 5 rolls), which has a probability of $(1-p)^5$ by independence.

(d) **(5 points)** What is the probability that on the Nth roll both dice show four?
(The random variable N is defined in part (c).)
We break this event up according to the value of N, and then use the independence of the trials.

$$\begin{aligned}
&P(\text{on the } N\text{th roll both dice show four}) \\
&= \sum_{k=1}^{\infty} P(\text{on the } k\text{th roll both dice show four, previous rolls have no fours}) \\
&= \sum_{k=1}^{\infty} (1-p)^{k-1} \frac{1}{6} \cdot \frac{1}{4} = \frac{1}{24} \sum_{k=1}^{\infty} (1-p)^{k-1} = \frac{1}{24} \cdot \frac{1}{1-(1-p)} = \frac{1}{24p} = \frac{1}{9}.
\end{aligned}$$

We also used the formula for the geometric series, and the fact that the probability of getting two fours when we roll the pair of dice is $P(A)P(B) = \frac{1}{24}$ (see part (a)).

(e) **(4 points)** Let N be defined as in part (c), and define the random variable Y as follows:

$$Y = \begin{cases} 1 & \text{if on the } N\text{th roll only the 6-sided die shows 4,} \\ 2 & \text{if on the } N\text{th roll only the 4-sided die shows 4,} \\ 3 & \text{if on the } N\text{th roll both dice show 4} \end{cases}$$

Decide whether N and Y are independent or not.

Solution: We will show that N and Y are independent, for this we need to check

$$P(N = k, Y = i) = P(N = k)P(Y = i)$$

for $k \geq 1$ and $1 \leq i \leq 3$.

We have $P(N = k) = p(1-p)^{k-1}$ since $N \sim \text{Geom}(p)$. We showed $P(Y = 3) = \frac{1}{24p}$ in part (c). Part of that computation was

$$P(N = k, Y = 3) = P(\text{on the } k\text{th roll both dice show four, previous rolls have no fours}) = (1-p)^{k-1}\frac{1}{24}$$

which gives

$$P(N = k, Y = 3) = (1-p)^{k-1}\frac{1}{24} = P(N = k)P(Y = 3) = p(1-p)^{k-1}\frac{1}{24p}.$$

Using the same approach we get

$$P(Y = 1) = \sum_{k=1}^{\infty}(1-p)^{k-1}\frac{1}{6}\cdot\frac{3}{4} = \frac{1}{8p}$$

and

$$P(N = k, Y = 1) = (1-p)^{k-1}\frac{1}{6}\cdot\frac{3}{4}$$

(this is the probability of not getting any fours for the first $k-1$ rolls and getting a four only on the 6-sided die on the kth roll). This shows that

$$P(N = k, Y = 1) = P(N = k)P(Y = 1)$$

Similarly,

$$P(Y = 2) = \sum_{k=1}^{\infty}(1-p)^{k-1}\frac{5}{6}\cdot\frac{1}{4} = \frac{5}{24p},$$

and

$$P(N = k, Y = 2) = (1-p)^{k-1}\frac{5}{6}\cdot\frac{1}{4}$$

showing

$$P(N = k, Y = 2) = P(N = k)P(Y = 2).$$

This shows that Y and N are indeed independent.

First Midterm Exam

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
Amy Cochran (MWF 9:55p)	Benedek Valkó (MWF 12:05p)	Sarah Tammen (TuTh 9:30a)	Shaoming Guo (MWF 8:50a)

- **There are 5 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total:	100	

1. **Short problems** The following short questions are not connected to each other. For each part, you have to *provide an appropriate explanation or justification.*

   (a) (5 points) Suppose that Y is a discrete random variable with a probability mass function given by

   $$p_Y(2) = \frac{1}{3}, \quad p_Y(3) = \frac{1}{6}, \quad p_Y(4) = \frac{1}{4}, \quad p_Y(7) = \frac{1}{4}.$$

   Find the probability that Y is even.

   (b) (5 points) In a lottery, 5 numbers are chosen uniformly at random from the numbers 1, 2, ..., 45. (We cannot choose the same number twice.) Find the probability that all 5 of the picked numbers are even. (*Do not simplify.*)

(Problem 1 continued)

(c) (5 points) The random variable X is continuous with probability density function given by

$$f(x) = \begin{cases} 1/2, & \text{if } x \in [-1, 0] \cup [2, 3] \\ 0, & \text{otherwise.} \end{cases}$$

Find $P(|X| > 1)$.

(d) (5 points) A, B, and C are events on the same probability space with $P(A) = 1/2$, $P(B) = 1/4$ and $P(C) = 1/8$. We know that A and B are independent, A and C are disjoint, and $C \subset B$. Find $P(A \cup B \cup C)$.

2. In a movie theater, there are 600 seats, arranged in 30 rows of 20 seats. Each ticket shows the row number and the seat number (e.g. ROW 21, SEAT 12). Before each showing, there is a random drawing among the 600 possible tickets (with each being equally likely), and the person sitting on the picked position (if there is someone there) gets their money back.

   (a) (8 points) Set up a probability space that models a *single* drawing. Make sure that you identify all components precisely.

   (b) (6 points) This weekend, the movie theater has 3 showings (with a random drawing before each one). Find the probability that the 3 picked tickets have 3 different row numbers.

(Problem 2 continued)

(c) (6 points) This month, the movie theater has 25 showings (with a random drawing before each one). Let X denote the number of drawn tickets among the 25 that are in the first row. Identify the distribution of X by name and specify the value of its parameter(s).

Note. If you do not remember the name, you can get full credit by writing down the probability mass function of X.

3. Bruno and Kevin are playing rock-paper-scissors. They both choose randomly one of the three objects, we may assume that all $3^2 = 9$ pairs are equally likely. If they choose different objects, then someone will win (rock beats scissors, scissors beat paper, paper beats rock). If they chose the same object (e.g. both chose paper), then they are tied, and they play again (by again choosing one of the objects randomly). In that case, they continue until someone wins.

   (a) (4 points) What is the probability that Bruno wins the game in the first round?

   (b) (6 points) What is the probability that the game ends in the fifth round, with Bruno using paper against Kevin's rock?

(Problem 3 continued)

(c) (6 points) What is the probability that the game ends with Bruno winning, with him choosing paper in the last game?

(d) (4 points) Let N denote the number of rounds they play until they find a winner. Identify the distribution of N by name with its parameter(s) and specify the value of its parameter(s).

Note. If you do not remember the name you can get full credit by writing down the probability mass function of N.

4. In a certain semester, the three sections of Math 431 have 35, 42, and 30 students enrolled. The three sections have 7, 12, and 9 second-year students (sophomores) enrolled, respectively. (So Section 2 has 42 students, out of which 12 are sophomores.)

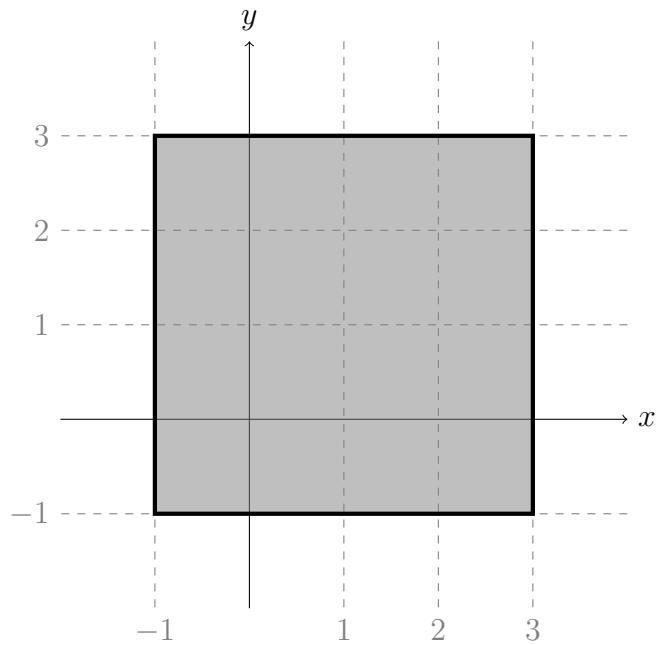
   (a) (6 points) We choose one of the three sections uniformly at random, and then we choose one of the students from that section (also uniformly at random). What is the probability that the chosen student is a sophomore? (*Do not simplify.*)

   (b) (8 points) We choose one of the three sections uniformly at random, and then we choose one of the students from that section (also uniformly at random). What is the probability that the chosen student is enrolled in the first section, given that they are a sophomore? (*Do not simplify.*)

(Problem 4 continued)

(c) (6 points) We choose a student uniformly at random among all the students enrolled in Math 431. Let X denote the section number of the student. Provide the probability mass function of X.

5. We choose a point (x, y) uniformly from the region $[-1, 3] \times [-1, 3]$ shaded in gray below.



(a) (6 points) Let X be the random variable that returns 1 if the x-value and y-value of the selected point share the same sign (i.e. they are both positive or both negative) and returns 0 otherwise. What is the probability of the event $\{X = 1\}$?

(Problem 5 continued)

(b) (7 points) Let A be the event that the selected point is in the region $[-1, 2] \times [-1, 0]$, and let B be the event that the selected point lies in the region $[-1, 0] \times [-1, c]$. Is it possible to choose $c > 0$ so that A and B are independent?

Hint. Draw a picture.

(Problem 5 continued)

(c) (7 points) Let D be the event that the selected point is in the region $[-1,3] \times [-1,0]$, and let E be the event that the selected point is in the region $[-1,0] \times [-1,3]$. Is it possible to find an event F with probability $P(F) = 1/2$, so that D, E and F are (mutually) independent?

Hint. The possible events can be represented as subsets of $[-1,3] \times [-1,3]$, but they do not have to be rectangular or even connected regions.

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First Midterm – Solutions

1. **Short problems**

(a) (5 points) Suppose that Y is a discrete random variable with a probability mass function given by

$$p_Y(2) = \frac{1}{3}, \quad p_Y(3) = \frac{1}{6}, \quad p_Y(4) = \frac{1}{4}, \quad p_Y(7) = \frac{1}{4}.$$

Find the probability that Y is even.

Solution: From the domain of the PMF we see that the possible values of Y are 2, 3, 4, and 7. Hence $\{Y \text{ is even}\} = \{Y \in \{2, 4\}\}$ and

$$P(Y \text{ is even}) = P(Y \in \{2, 4\}) = P(Y = 2) + P(Y = 4) = p_Y(2) + p_Y(4) = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}.$$

(b) (5 points) In a lottery, 5 numbers are chosen uniformly at random from the numbers 1, 2, ..., 45. (We cannot choose the same number twice.) Find the probability that all 5 of the picked numbers are even. (*Do not simplify.*)

Solution: This is sampling without replacement from the set $\{1, \ldots, 45\}$ with a sample of size 5. Consider the sample space with order not recorded, then $\#\Omega = \binom{45}{5}$ (the number of size 5 subsets of $\{1, \ldots, 45\}$). Let A be the event that all 5 picked numbers are even. There are 22 even numbers between 1 and 45, and we can choose $\binom{22}{5}$ subsets of size 5 from these numbers. Hence

$$P(A) = \frac{\#A}{\#\Omega} = \frac{\binom{22}{5}}{\binom{45}{5}} = \frac{22 \cdot 21 \cdot 20 \cdot 19 \cdot 18}{45 \cdot 44 \cdot 43 \cdot 42 \cdot 41}$$

(c) (5 points) The random variable X is continuous with probability density function given by

$$f(x) = \begin{cases} 1/2, & \text{if } x \in [-1, 0] \cup [2, 3] \\ 0, & \text{otherwise.} \end{cases}$$

Find $P(|X| > 1)$.

Solution: The event $\{|X| > 1)\}$ is the same as $\{X \in (-\infty, -1] \cup [1, \infty)\}$. Since X is continuous, we can compute $P(|X| \geq 1)$ by integrating the pdf on the set $(-\infty, -1] \cup [1, \infty)$:

$$P(|X| \geq 1) = \int_{-\infty}^{-1} f(x)dx + \int_{1}^{\infty} f(x)dx$$

On the set $(-\infty, -1)$ the pdf is equal to 0 (we don't have to worry about the value at -1), so $\int_{-\infty}^{-1} f(x)dx = 0$.

On the set $(1, \infty)$ the pdf is equal to $1/2$ on $[2, 3]$, and it is zero otherwise, so

$$\int_{1}^{\infty} f(x)dx = \int_{2}^{3} \frac{1}{2}dx = \frac{1}{2}.$$

Hence $P(|X| \geq 1) = \frac{1}{2}$.

(d) (5 points) A, B, and C are events on the same probability space with $P(A) = 1/2$, $P(B) = 1/4$ and $P(C) = 1/8$. We know that A and B are independent, A and C are disjoint, and $C \subset B$. Find $P(A \cup B \cup C)$.

Solution: We can compute $P(A \cup B \cup C)$ using inclusion-exclusion:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC).$$

We know that $P(A) = 1/2$, $P(B) = 1/4$, $P(C) = 1/8$. Since A and B are independent, we have

$$P(AB) = P(A)P(B) = \frac{1}{8}.$$

Since $C \subset B$, we have $BC = C$, and $P(BC) = \frac{1}{8}$. Finally, since A and C are independent, we have $AC = \emptyset$ and $P(AC) = 0$. This also shows that $ABC = \emptyset$ and $P(ABC) = 0$. Plugging everything into our equation gives

$$P(A \cup B \cup C) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} - \frac{1}{8} - \frac{1}{8} = \frac{5}{8}.$$

Here is another way to get this answer. Since $C \subset B$, we have $B \cup C = B$ and $A \cup B \cup C = A \cup B$. Now we can use inclusion-exclusion for just two events to get

$$P(A \cup B \cup C) = P(A \cup B) = P(A) + P(B) - P(AB) = P(A) + P(B) - P(A)P(B) = \frac{1}{2} + \frac{1}{4} - \frac{1}{8} = \frac{5}{8}.$$

2. In a movie theater, there are 600 seats, arranged in 30 rows of 20 seats. Each ticket shows the row number and the seat number (e.g. ROW 21, SEAT 12). Before each showing, there is a random drawing among the 600 possible tickets (with each being equally likely), and the person sitting on the picked position (if there is someone there) gets their money back.

(a) (8 points) Set up a probability space that models a *single* drawing. Make sure that you identify all components precisely.

Solution: We can encode each ticket with a pair of integer numbers (r, s) where $1 \leq r \leq 30$ denotes the row and $1 \leq s \leq 10$ denotes the seat number. Our sample space is the collection of these pairs:

$$\Omega = \{(r, s) : 1 \leq r \leq 30, 1 \leq s \leq 10, \ r, s \text{ are integers}\}$$

Note that $\#\Omega = 30 \cdot 20 = 600$. Our set of events is all the subsets of Ω. Because the outcomes are equally likely, for each event A we have

$$P(A) = \frac{\#A}{\#\Omega} = \frac{\#A}{600}.$$

(b) (6 points) This weekend, the movie theater has 3 showings (with a random drawing before each one). Find the probability that that the 3 picked tickets have 3 different row numbers.

Solution: This is sampling with replacement with a sample of size 3 from the set Ω from part (a). Our new sample space $\widetilde{\Omega}$ is just the set of triples from Ω:

$$\widetilde{\Omega} = \Omega^3 = \{(r_1, s_2, r_2, s_2, r_3, s_3), 1 \leq r_i \leq 30, 1 \leq s_i \leq 20\}$$

(and r_i, s_i are integers for $i = 1, 2, 3$). The size of $\widetilde{\Omega}$ is $(\#\Omega)^3 = 600^3$.

Let B be the event that the three row numbers r_1, r_2, r_3 are different. We can choose the row numbers $30 \cdot 29 \cdot 28$ different ways: r_1 can take 30 values, once we picked it r_2 cannot take the same value, so

it can only take 29 possible values, and once $r_1 \neq r_2$ are picked we have 28 choices for r_3. We do not have any restrictions on the three seat numbers, so those can be picked 20^3 different ways. This gives $\#B = 30 \cdot 29 \cdot 28 \cdot 20^3$ and

$$P(B) = \frac{30 \cdot 29 \cdot 28 \cdot 20^3}{(30 \cdot 20)^3} = \frac{30 \cdot 29 \cdot 28}{30^3} = \frac{29 \cdot 28}{30^2}.$$

(c) (6 points) This month, the movie theater has 25 showings (with a random drawing before each one). Let X denote the number of drawn tickets among the 25 that are in the first row. Identify the distribution of X by name and specify the value of its parameter(s).

Note. If you do not remember the name, you can get full credit by writing down the probability mass function of X.

Solution: Let us call a drawing successful if the drawn ticket us in the first row. Since the drawings are independent, we can treat X as the number of successes within 25 independent trials with the same success probability, so it will have binomial distribution.

The probability of a success is $\frac{20}{600} = \frac{1}{30}$ (there are 20 seats in the first row). There are 25 trials so $X \sim \text{Bin}(25, \frac{1}{30})$.

The probability mass function is

$$P(X = k) = \binom{25}{k} \left(\frac{1}{30}\right)^k \left(\frac{29}{30}\right)^{25-k}, \qquad k = 0, 1, \ldots, 25.$$

3. Bruno and Kevin are playing rock-paper-scissors. They both choose randomly one of the three objects, we may assume that all $3^2 = 9$ pairs are equally likely. If they choose different objects, then someone will win (rock beats scissors, scissors beat paper, paper beats rock). If they chose the same object (e.g. both chose paper) then they are tied, and they play again (by again choosing one of the objects randomly). In that case they continue until someone wins.

(a) (4 points) What is the probability that Bruno wins the game in the first round?

Solution: We are choosing 1 out of the $3^2 = 9$ pairs, and each one is equally likely. Three of those pairs result in Bruno winning: rock beats scissors, scissors beat paper, paper beats rock. Therefore the probability that Bruno wins in the first round is $3/9$.

Note that 3 of the 9 pairs give ties (when both players choose the same object), and 3 of the pairs result in a win for Kevin. These also have $1/3 - 1/3$ probabilities.

(b) (6 points) What is the probability that the game ends in the fifth round, with Bruno using paper against Kevin's rock?

Solution: If the game ends in the fifth round, it means the first four rounds are all ties. In each round, the probability of having a tie is $3/9 = 1/3$. In the fifth round, the probability that Bruno uses paper against Kevin's rock is $1/9$ (this is one of the 9 equally likely pairs). Because of independence, the probability is

$$\left(\frac{1}{3}\right)^4 \times \frac{1}{9} = \frac{1}{729}.$$

(c) (6 points) What is the probability that the game ends with Bruno winning, with him choosing paper in the last game?

Solution: Let A be the event that Bruno wins the game with choosing paper in the last game, and let $A_k, k = 1, 2, \ldots$ be the event that Bruno wins the game in the kth game with choosing paper in that game. Then $A = \cup_{k=1}^{\infty} A_k$, the events $A_1, A_2, \ldots$ are disjoint, hence by additivity

$$P(A) = \sum_{k=1}^{\infty} P(A_k).$$

In part (b) we computed $P(A_5)$, the same argument can be used to compute $P(A_k)$ for a general k:

$$P(A_k) = \left(\frac{1}{3}\right)^{k-1} \times \frac{1}{9} = \frac{1}{3^{k+1}}.$$

This leads to

$$P(A) = \sum_{k=1}^{\infty} P(A_k) = \sum_{k=1}^{\infty} \frac{1}{3^{k+1}} = \frac{1}{9} \cdot \frac{1}{1 - 1/3} = \frac{1}{6},$$

where we used the geometric series formula.

Alternate solution: Here is a slick way to get the same answer using symmetry (without using the geometric series). We know that the game ends with probability one (this can be shown similarly as we showed that eventually we flip heads with a fair coin). If the game ends, then we can have the following six outcomes in the final game: Bruno wins with choosing paper in the last game, Bruno wins with choosing rock in the last game, Bruno wins with choosing scissors in the last game, Kevin wins with choosing paper in the last game, Kevin wins with choosing rock in the last game, Kevin wins with choosing scissors in the last game. These six events should all have the same probability (this follows from symmetry, for the precise proof you would need to divide these events further based on how many rounds were played), and the sum of the probabilities is 1, so each will be equal to 1/6.

(d) (4 points) Let N denote the number of rounds they play until they find a winner. Identify the distribution of N by name with its parameter(s) and specify the value of its parameter(s).

Note. If you do not remember the name you can get full credit by writing down the probability mass function of N.

Solution: We can treat each round as an independent trial, where getting a tie is a failure, and one of the players winning means a success. Then N just counts the number of trials until we see the first success, so this is geometric distribution. The success probability is $1 - 1/3 = 2/3$, so $N \sim \text{Geo}(2/3)$. The probability mass function is given by

$$P(N = k) = \left(\frac{1}{3}\right)^{k-1} \left(\frac{2}{3}\right),$$

where $k = 1, 2, \ldots$

4. In a certain semester, the three sections of Math 431 have 35, 42, and 30 students enrolled. The three sections have 7, 12, and 9 second-year students (sophomores) enrolled, respectively. (So Section 2 has 42 students, out of which 12 are sophomores.)

(a) (6 points) We choose one of the three sections uniformly at random, and then we choose one of the students from that section (also uniformly at random). What is the probability that the chosen student is a sophomore? (*Do not simplify.*)

Solution: Let A be the event that the chosen student is a sophomore, and let B_k be the event that the chosen section is section k with $k = 1, 2, 3$. Note that B_1, B_2, B_3 forms a partition of our sample space. We are interested in A, which we can calculate using the law of total probability (i.e. conditioning on which section is chosen):

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3).$$

Because we choose sections uniformly at random,

$$P(B_1) = P(B_2) = P(B_3) = 1/3.$$

Similarly, because we choose students uniformly at random upon selecting a section, then $P(A|B_k)$ is the number of sophomores in Section k divided by the total number of students in section k. Putting this together, we get

$$P(A) = \frac{7}{35} \times \frac{1}{3} + \frac{12}{42} \times \frac{1}{3} + \frac{9}{30} \times \frac{1}{3} = \frac{11}{42}.$$

(b) (8 points) We choose one of the three sections uniformly at random, and then we choose one of the students from that section (also uniformly at random). What is the probability that the chosen student is enrolled in the first section, given that they are a sophomore? (*Do not simplify.*)

Solution: Using the events defined in part (a), we are interested in $P(B_1|A)$. We can compute this using Bayes' formula:

$$P(B_1|A) = \frac{P(A|B_1)P(B_1)}{P(A)}.$$

Using our answers from (a), we have

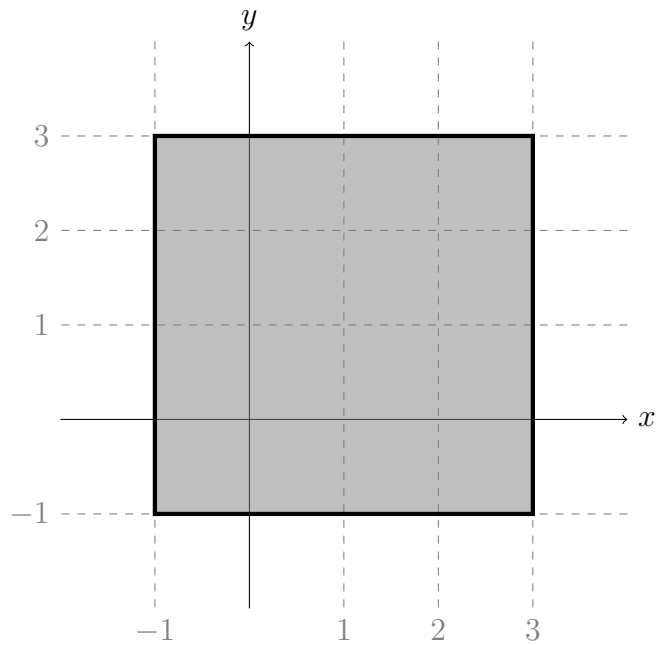
$$P(B_1|A) = \frac{\frac{7}{35} \times \frac{1}{3}}{\frac{7}{35} \times \frac{1}{3} + \frac{12}{42} \times \frac{1}{3} + \frac{9}{30} \times \frac{1}{3}} = \frac{\frac{7}{35} \times \frac{1}{3}}{\frac{11}{42}} = \frac{14}{55}.$$

(c) (6 points) We choose a student uniformly at random among all the students enrolled in Math 431. Let X denote the section number of the student. Provide the probability mass function of X.

Solution: The random variable X can take three values: 1, 2, or 3. Since students are chosen uniformly at random, then the probability that $X = k$ is simply the number of students in Section k divided by the total number of students. There are 107 students in total. Hence, the probability mass function of X is given by:

$$p_X(k) = \begin{cases} \frac{35}{107} & k = 1 \\ \frac{42}{107} & k = 2 \\ \frac{30}{107} & k = 3 \end{cases}.$$

5. We choose a point (x, y) uniformly from the region $[-1, 3] \times [-1, 3]$ shaded in gray below.



(a) (6 points) Let X be the random variable that returns 1 if the x-value and y-value of the selected point share the same sign (i.e. they are both positive or both negative) and returns 0 otherwise. What is the probability of the event $\{X = 1\}$?

Solution: The event $\{X = 1\}$ is the set of points (x, y) inside the square $[-1, 3] \times [-1, 3]$ with $x > 0, y > 0$ or $x < 0, y < 0$. This set is just the union of the squares $(0, 3] \times (0, 3]$ and $[-1, 0) \times [-1, 0)$, so

$$\{X = 1\} = (0, 3] \times (0, 3] \cup [-1, 0) \times [-1, 0)$$

In this probability space we can compute the probability of a subset A of $\Omega = [-1, 3] \times [-1, 3]$ with the formula $\frac{\text{area}(A)}{\text{area}(\Omega)}$, so

$$P(X = 1) = P((0, 3] \times (0, 3]) + P([-1, 0) \times [-1, 0)) = \frac{9}{16} + \frac{1}{16} = \frac{5}{8}.$$

(b) (7 points) Let A be the event that the selected point is in the region $[-1, 2] \times [-1, 0]$, and let B be the event that the selected point lies in the region $[-1, 0] \times [-1, c]$. Is it possible to choose $c > 0$ so that A and B are independent?

Solution: A and B are independent exactly if $P(A)P(B) = P(AB)$. We have

$$P(A) = \frac{\text{area}([-1, 2] \times [-1, 0])}{16} = \frac{3}{16}.$$

Since $c > 0$, the intersection AB is the square $[-1, 0] \times [-1, 0]$, so

$$P(AB) = \frac{\text{area}([-1, 0] \times [-1, 0]}{16} = \frac{1}{16}.$$

For the independence we need

$$P(A)P(B) = \frac{3}{16} \cdot P(B) = \frac{1}{16} = P(AB)$$

which would lead to

$$P(B) = 1/3.$$

That means that the area of B should be $16/3$. Since B is a rectangle where one side has length 1, the other side should be $16/3 > 4$. But this is impossible: that side cannot be longer than the corresponding side of the square $[-1,3] \times [-1,3]$. Hence we cannot choose a $c > 0$ so that A and B are independent.

(c) (7 points) Let D be the event that the selected point is in the region $[-1,3] \times [-1,0]$, and let E be the event that the selected point is in the region $[-1,0] \times [-1,3]$. Is it possible to find an event F with probability $P(F) = 1/2$, so that D, E and F are (mutually) independent?

Hint. The possible events can be represented as subsets of $[-1,3] \times [-1,3]$, but they do not have to be rectangular or even connected regions.

Solution: By computing the respective areas we get $P(D) = P(E) = \frac{1}{4}$ and $P(DE) = \frac{1}{16}$. We also get that $DE = [-1,0] \times [-1,0]$. (Hence D and E are independent.)

If D, E, and F are mutually independent then we must have

$$P(DE) = P(D)P(E), \quad P(D)P(F) = P(DF), \quad P(EF) = P(E)P(F), \quad P(DEF) = P(D)P(E)P(F).$$

We have already checked that the first equation holds. Using $P(D) = P(E) = 1/4$ and $P(F) = 1/2$ the other three equations are equivalent to

$$P(DF) = \frac{1}{8}, \quad P(EF) = \frac{1}{8}, \quad P(DEF) = \frac{1}{32}.$$

which means that

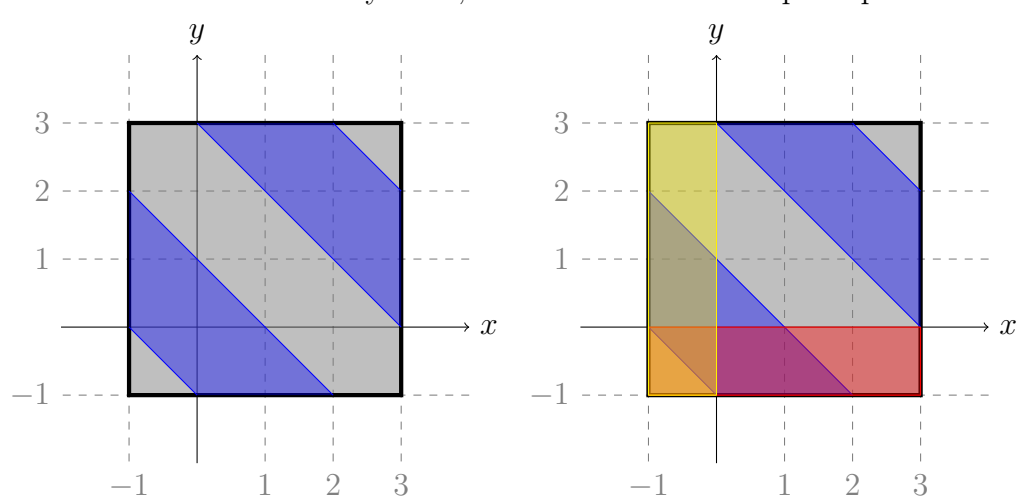
$$\text{area}(DF) = 2, \qquad \text{area}(EF) = 2, \qquad \text{area}(DEF) = 1/2,$$

and $\text{area}(F) = 8$.

There are a number of ways we can create a region F satisfying these conditions. One example of a valid F is given by

$$F = \{(x,y) \in [-1,3]^2 : -1 \le x + y \le 1\} \cup \{(x,y) \in [-1,3]^2 : -1 \le x + y \le 1\} =: F_1 \cup F_2.$$

This set is shown below in blue - first by itself, then with D and E superimposed.



Each of the components F_1 and F_2 is a trapezoid of area 2; we can see that the area of each trapezoid is 4 either by counting half-squares or by noticing that each trapezoid consists of a right triangle with a smaller right triangle cut out. For instance, to form F_1, we can start with the triangle T_1 whose vertices are at $(-1,1)$, $(-1,2)$, and $(2,-1)$ and then remove the smaller triangle whose vertices are at $(-1,-1)$, $(-1,0)$, and $(0,-1)$. This means that

$$\text{area}(F_1) = \frac{1}{2}(3)(3) - \frac{1}{2}(1)(1) = \frac{1}{2}(8) = 4.$$

Since F_2 is a congruent copy of F_1 which does not intersect F_1, we have that

$$\text{area}(F) = \text{area}(F_1) + \text{area}(F_2) = 8.$$

Meanwhile, each of DF and EF is a parallelogram of base 2 and height 1; thus,

$$\text{area}(DF) = 2 = \text{area}(EF).$$

Finally, the intersection DEF is a small right triangle whose legs have length 1; thus,

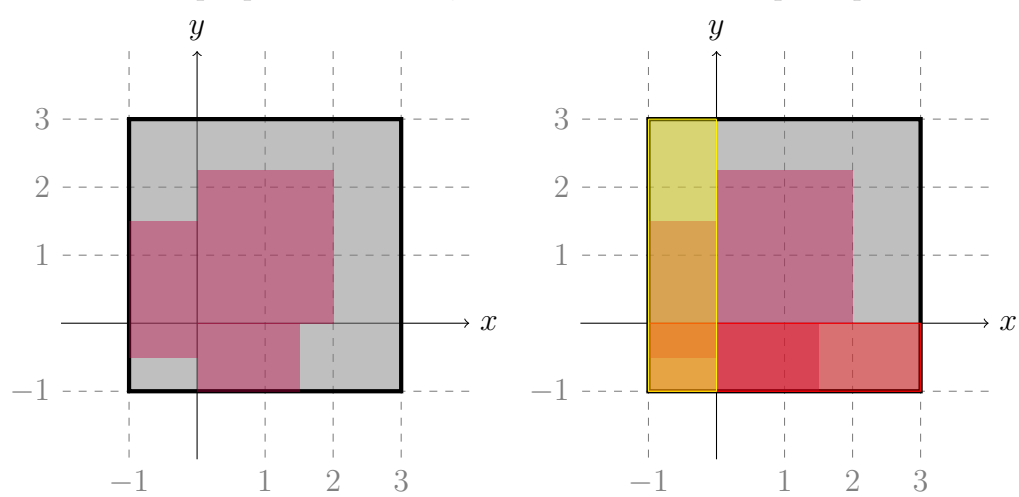
$$\text{area}(DEF) = \frac{1}{2},$$

as desired.

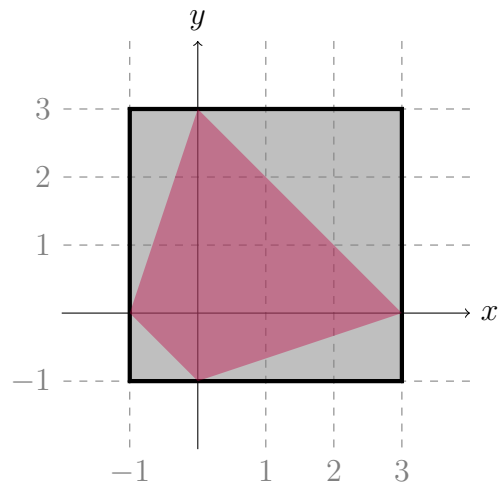
Another example of a valid F is the union

$$F = [-1,0] \times [-1/2,0] \cup [-1,0] \times [0,3/2] \cup [0,3/2] \times [-1,0] \cup [0,2] \times [0,9/4].$$

This set is shown below in purple - first alone, then with D and E superimposed.



Here is one more possible solution for F (see picture on the next page): this is the trapezoid with vertices $(-1,0)$, $(0,-1)$, $(3,0)$, $(0,3)$. This region contains exactly half of each of the sets DE, DE^c, D^cE, D^cE^c. Using the alternative description of the independence of multiple events (Fact 2.23 from the textbook), and the fact that D and E are independent, it follows that D, E, F are independent as well.


y
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2
3

Department of Mathematics, University of Wisconsin-Madison

Math 431 — Midterm Exam 1 — Fall 2024

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: David Clancy or Mikhail Ivanov

Time: **90 minutes**

- This exam contains 6 questions some with multiple parts, 10 pages (including the cover) for the total of 93 marks. Read the problems carefully and budget your time wisely.
- You are allowed a single sheet of hand-written notes.
- **NO CALCULATORS** or other electronic devices are to be used. Turn off your phone so as to not disturb others.
- You **do not** need to simplify binomial coefficients or factorials.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded.
- Justify your steps.
- If you use an additional page for a particular problem, be sure to **CLEARLY** indicate this on the problem's page so I know to look further.

Question:	1	2	3	4	5	6	Total
Points:	16	15	16	26	10	10	93
Score:							

1. (16 points) The following questions are **Answers Only**.

   Every morning, Albert flips five fair (5) coins. He calls a day good if at least four (4) coins are heads.

   a) What is the probability that today (October 9th) was a good day? Call this number p.

   b) October has 31 days. Let X be the number of good days Albert has in October. What is the distribution of X? Either name it or write down explicitly the probability mass function.

   c) September 30th was a Monday and a good day for Albert. What is the probability the first good day after September 30th was Monday October 7th?

   d) What is the probability that the first good day *after* September 30th is also on a Monday? You can leave your answer in terms of the number p you computed in part (a).

2. You have one fair coin, and one biased coin which lands Heads with probability 3/4.

   (a) (5 points) You pick one of the coins uniformly at random and flip it **once**. what is the probability that the coin you picked lands Head?

   (b) (10 points) You pick one of the coins at random and flip it **three times**. It lands Heads all three times. Given this information, what is the probability that the coin you picked is the fair one?

3. The following questions are **Answers Only**.

(a) (4 points) Let X be a random variable with cumulative distribution function

$$F(x) = \begin{cases} \frac{1}{2}e^x & : x \leq 0 \\ \frac{1}{2} + \frac{1}{4}x & : 0 \leq x \leq 2 \\ 1 & : x \geq 2. \end{cases}$$

What is $P(-1 < X \leq 1)$?

(b) (4 points) A and B are independent events such that $P(A) = 0.44$ and $P(B) = \frac{1}{4} = 0.25$. What is $P(A \cup B)$?

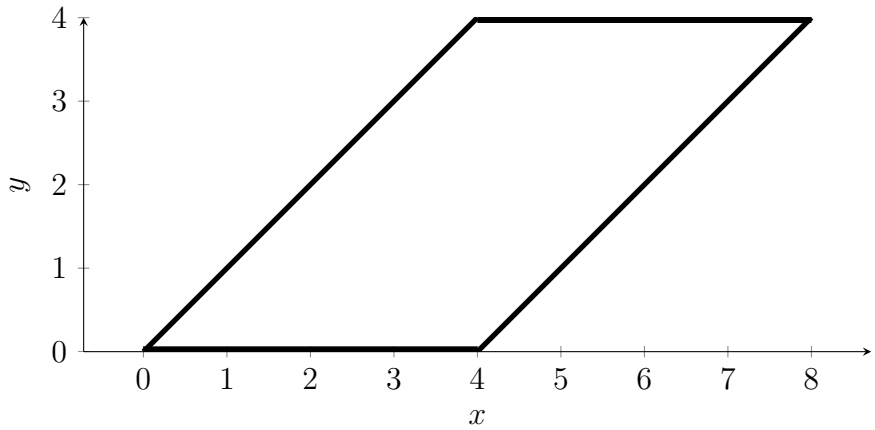
(c) (4 points) An urn contains 17 red balls, 13 green balls, and 4 yellow balls. Sample 15 **without replacement**. Let X be the number of green balls drawn. What is the distribution of X? Either name it (with parameters) or write down its probability mass function.

(d) (4 points) Let X be a random variable with probability mass function

$$P(X = 0) = \frac{1}{3} \qquad \text{and} \qquad P(X = n) = c\left(\frac{1}{4}\right)^n \text{ for } n = 1, 2, \ldots.$$

What is the value of c?

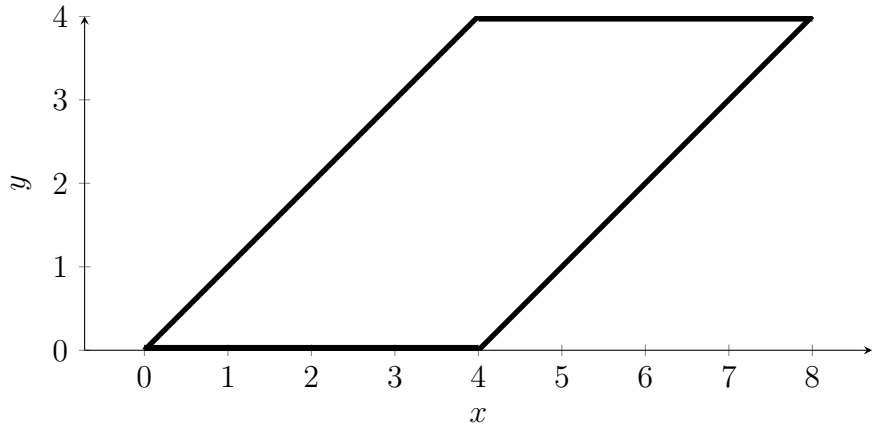
4. Sample a point (X, Y) uniformly at random inside the parallelogram with vertices $(0, 0)$, $(4, 0)$, $(8, 4)$, and $(4, 4)$. This is depicted in figure below (and copied on the second page of this problem).



(a) (4 points) What is $P(Y \leq 3)$?

(b) (4 points) What is $P(X \leq 4, Y \leq 3)$?

(c) (4 points) Are X and Y independent? Explain your answer.



(d) (7 points) What is the cumulative distribution function of X?

(e) (7 points) What is the probability density function of X?

5. (10 points) Suppose that in the population of college applicants, being good at baseball is independent of having a good math score on a certain standardized test. A certain college has a simple admissions procedure: admit an applicant if and only if the applicant is good at baseball or has a good math score on the test. Let's show why it makes sense that among students that the college admits, having a good math score decreases their likelihood of being good at baseball, i.e., conditioning on having a good math score decreases the chance of being good at baseball.

   Mathematically speaking show that if A and B are independent with $P(A) = 3/4$ and $P(B) = 1/3$ and $C = A \cup B$, then A and B are conditionally dependent given C, with

$$P(A|B, C) < P(A|C).$$

   where $P(A|B, C) = P(A|BC) = P(A|B \cap C)$.

6. (10 points) Bob is playing a video game that has 7 levels. He starts at level 1, and has probability 0.2 of reaching level 2. In general, given that he reaches level j, he has probability 0.2 of reaching level $j+1$, for $1 \leq j \leq 6$. Let X be the highest level that he reaches. Find the PMF of X.

SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM.
SCRATCH WORK WILL NOT BE GRADED

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Department of Mathematics, University of Wisconsin-Madison

Math 431 — Midterm Exam 1 — Solutions — Fall 2024

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: David Clancy or Mikhail Ivanov

Time: **90 minutes**

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- You **do not** need to simplify binomial coefficients or factorials.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded.
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Question:	1	2	3	4	5	6	Total
Points:	16	15	16	26	10	10	93
Score:							

1. (16 points) The following questions are **Answers Only**.

Every morning, Albert flips five fair (5) coins. He calls a day good if at least four (4) coins are heads.

a) What is the probability that today (October 9th) was a good day? Call this number p.

Solution: The number of heads on any day is $\text{Bin}(n = 5, p = 1/2)$ so

$$P(\geq 4 \text{ heads}) = \binom{5}{4}\left(\frac{1}{2}\right)^4 \frac{1}{2} + \binom{5}{5}\left(\frac{1}{2}\right)^5 = \frac{5+1}{2^5} = \frac{3}{16}.$$

b) October has 31 days. Let X be the number of good days Albert has in October. What is the distribution of X? Either name it or write down explicitly the probability mass function.

Solution: Each day is good independently with probability $p = \frac{3}{16}$ (from part (a)). There are $n = 31$ days in October. So $X \sim \text{Bin}(n = 31, p = \frac{3}{16})$.
Its probability mass function is

$$p_X(k) = \binom{31}{k} p^k (1-p)^{31-k} \qquad \text{for } k = 0, 1, \cdots, 31.$$

c) September 30th was a Monday and a good day for Albert. What is the probability the first good day after September 30th was Monday October 7th?

Solution: Let N be the number of days after September 30th for Albert to have another good day. Then $N \sim \text{Geom}(p)$ where p is as in part (a). Then

$$P(N = 7) = (1-p)^{7-1}p = (1-p)^6 p.$$

d) What is the probability that the first good day *after* September 30th is also on a Monday? You can leave your answer in terms of the number p you computed in part (a).

Solution: Let N be as above. We want to compute

$$\begin{aligned}
P(N = 7, 14, 21, \cdots) &= \sum_{n=1}^{\infty} P(N = 7n) \\
&= \sum_{n=1}^{\infty} (1-p)^{7n-1} p \\
&= \frac{p}{1-p} \sum_{n=1}^{\infty} (1-p)^{7n} \\
&= \frac{p}{1-p} \sum_{n=1}^{\infty} r^n \qquad \text{where } r = (1-p)^7 \\
&= \frac{p}{1-p} \cdot \frac{r}{1-r} = \frac{p}{1-p} \cdot \frac{(1-p)^7}{1-(1-p)^7} = \frac{p(1-p)^6}{1-(1-p)^7}.
\end{aligned}$$

2. You have one fair coin, and one biased coin which lands Heads with probability 3/4.

   (a) (5 points) You pick one of the coins uniformly at random and flip it **once**. what is the probability that the coin you picked lands Head?

Solution: Let H be the event you flip heads, F the event you flip the fair coin and B the biased coin. Then, by law of total probability

$$P(H) = P(H|F)P(F) + P(H|B)P(B) = \frac{1}{2} \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8}.$$

   (b) (10 points) You pick one of the coins at random and flip it **three times**. It lands Heads all three times. Given this information, what is the probability that the coin you picked is the fair one?

Solution: Let H_3 be the event that we flip heads 3 times. Let B and F be the same as before. We want

$$P(F|H_3) = \frac{P(H_3|F)P(F)}{P(H_3)}.$$

We compute

$$P(H_3 \cap F) = P(H_3|F)P(F) = \left(\frac{1}{2}\right)^3 \frac{1}{2} = \frac{1}{16} = \frac{8}{128}$$

and

$$P(H_3 \cap B) = P(H_3|B)P(B) = \left(\frac{3}{4}\right)^3 \frac{1}{2} = \frac{27}{128}.$$

So $P(H_3) = P(H_3 \cap F) + P(H_3 \cap B) = \frac{27}{128} + \frac{8}{128} = \frac{35}{128}$. Now

$$P(F|H_3) = \frac{P(H_3|F)P(F)}{P(H_3)} = \frac{8/128}{35/128} = \frac{8}{35}.$$

3. The following questions are **Answers Only**.

(a) (4 points) Let X be a random variable with cumulative distribution function

$$F(x) = \begin{cases} \frac{1}{2}e^x & : x \leq 0 \\ \frac{1}{2} + \frac{1}{4}x & : 0 \leq x \leq 2 \\ 1 & : x \geq 2. \end{cases}$$

What is $P(-1 < X \leq 1)$?

Solution: $F(1) - F(-1) = \frac{3}{4} - \frac{1}{2}e^{-1}$.

(b) (4 points) A and B are independent events such that $P(A) = 0.44$ and $P(B) = \frac{1}{4} = 0.25$. What is $P(A \cup B)$?

Solution: $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = .44 + .25 - .11 = 0.58$.

(c) (4 points) An urn contains 17 red balls, 13 green balls, and 4 yellow balls. Sample 15 **without replacement**. Let X be the number of green balls drawn. What is the distribution of X? Either name it (with parameters) or write down its probability mass function.

Solution: X is a hypergeometric random variable with parameters $N = 17 + 13 + 4 = 34$ (total balls), $N_A = 13$ (green balls), and $n = 15$ (drawn balls). So $X \sim \text{HyperGeom}(34, 13, 15)$.

Alternatively,

$$P(X = k) = \frac{\binom{13}{k}\binom{21}{15-k}}{\binom{34}{15}} \qquad \text{for } k = 0, 1, 2, \ldots, 13, \quad \text{and 0 otherwise.}$$

(d) (4 points) Let X be a random variable with probability mass function

$$P(X = 0) = \frac{1}{3} \qquad \text{and} \qquad P(X = n) = c\left(\frac{1}{4}\right)^n \text{ for } n = 1, 2, \ldots.$$

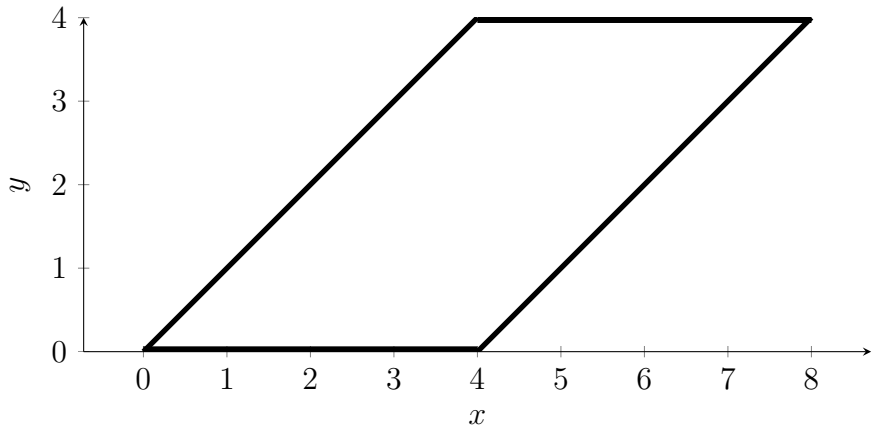
What is the value of c?

Solution: We need

$$1 = \sum_{n=0}^{\infty} P(X = n) = \frac{1}{3} + \sum_{n=1}^{\infty} c\frac{1}{4^n} = \frac{1}{3} + c\frac{1/4}{1 - 1/4} = \frac{1}{3} + \frac{c}{3}$$

so $c = 2$.

4. Sample a point (X, Y) uniformly at random inside the parallelogram with vertices $(0, 0)$, $(4, 0)$, $(8, 4)$, and $(4, 4)$. This is depicted in figure below (and copied on the second page of this problem).



(a) (4 points) What is $P(Y \leq 3)$?

Solution: For $Y \leq 3$, the point (X, Y) needs to be within the parallelogram with vertices $(0, 0), (4, 0), (7, 3), (3, 3)$. This has area $12 = 3 \times 4$. The total area of Ω is $16 = 4 \times 4$. Hence $P(Y \leq 3) = \frac{12}{16} = \frac{3}{4}$.

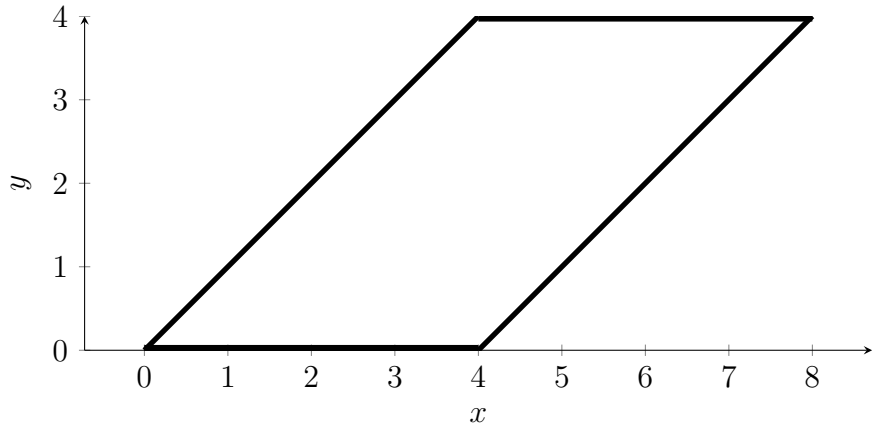
(b) (4 points) What is $P(X \leq 4, Y \leq 3)$?

Solution: We now wan the trapezoid with vertices $(0, 0), (4, 0), (4, 3), (3, 3)$ which has area $\frac{1+4}{2} \times 3 = 15/2$. Hence $P(X \leq 4, Y \leq 3) = (15/2)/16 = 15/32$.

(c) (4 points) Are X and Y independent? Explain your answer.

Solution: No. The probability $P(X \leq 4) = \frac{1}{2}$ and

$$P(X \leq 4)P(Y \leq 3) = \frac{1}{2}\frac{3}{4} = \frac{3}{8} \neq \frac{15}{32} = P(X \leq 4, Y \leq 3).$$

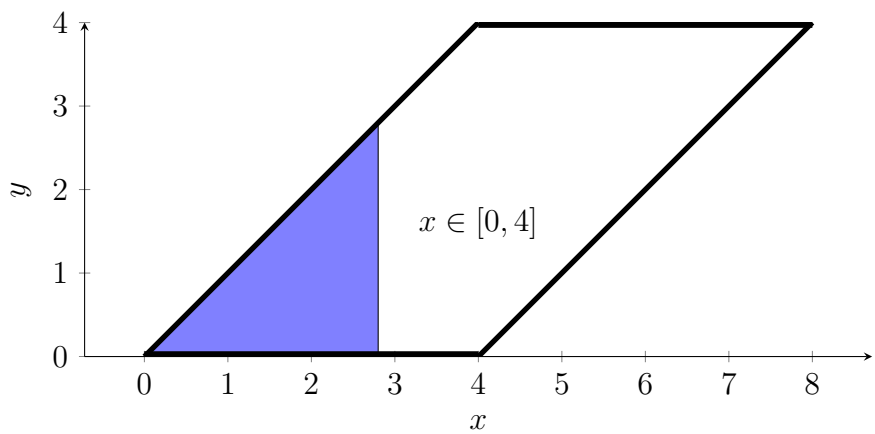


(d) (7 points) What is the cumulative distribution function of X?

Solution: The area of the parallelogram is 16. For $x \leq 0$, $F_X(x) = 0$. For $x \in [0, 4]$ we can compute

$$F_X(x) = \frac{\text{area of triangle with height } x \text{ and base } x}{\text{area of parallelogram}} = \frac{x^2}{32}.$$

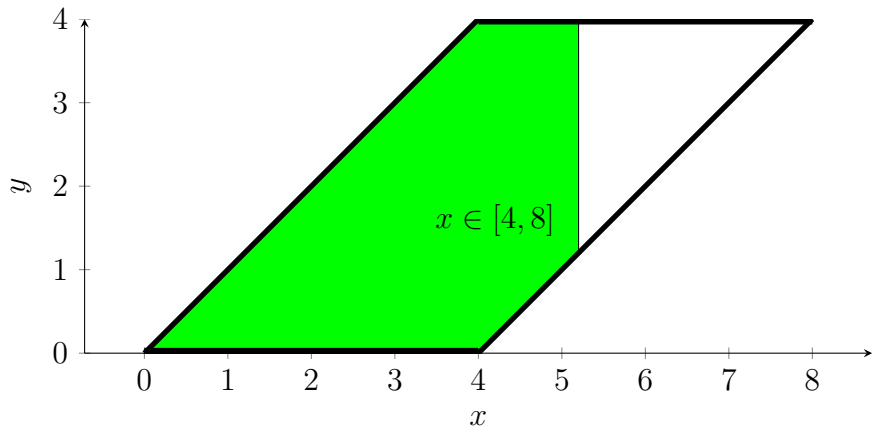
This is in the figure below.



For $x \in [4, 8]$ it is easier to compute the complement probability.

$$F_X(x) = 1 - \frac{\text{area of triangle with base } 8 - x \text{ and height } (8 - x)}{16} = 1 - \frac{(8 - x)^2}{32}.$$

This is the picture below.



Lastly, $F_X(x) = 1$ for all $x \geq 4$. Combining everything:

$$F_X(x) = \begin{cases} 0 & : x \leq 0 \\ \frac{x^2}{32} & : 0 \leq x \leq 4 \\ 1 - \frac{(8-x)^2}{32} = \frac{-x^2+16x-32}{32} & : 4 \leq x \leq 8 \\ 1 & : x \geq 4 \end{cases}.$$

(e) (7 points) What is the probability density function of X?

Solution: We differentiate:

$$F'_X(x) = \begin{cases} 0 & : x \leq 0 \\ \frac{x}{16} & : 0 \leq x \leq 4 \\ -\frac{x}{16} + \frac{1}{2} & : 4 \leq x \leq 8 \\ 0 & : x \geq 4 \end{cases}.$$

5. (10 points) Suppose that in the population of college applicants, being good at baseball is independent of having a good math score on a certain standardized test. A certain college has a simple admissions procedure: admit an applicant if and only if the applicant is good at baseball or has a good math score on the test. Let's show why it makes sense that among students that the college admits, having a good math score decreases their likelihood of being good at baseball, i.e., conditioning on having a good math score decreases the chance of being good at baseball.

   Mathematically speaking show that if A and B are independent with $P(A) = 3/4$ and $P(B) = 1/3$ and $C = A \cup B$, then A and B are conditionally dependent given C, with

$$P(A|B, C) < P(A|C).$$

   where $P(A|B, C) = P(A|BC) = P(A|B \cap C)$.

Solution: First, $B \cap C = B$ so

$$P(A|B, C) = P(A|B \cap C) = \frac{P(A \cap (B \cap C))}{P(B \cap C)} = \frac{P(A \cap B)}{P(B)}$$

and since A, B are independent

$$P(A \cap B) = P(A)P(B).$$

Hence $P(A|B, C) = P(A) = \frac{3}{4}$.

Second,

$$P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{P(A)}{P(C)} = \frac{P(A)}{P(A \cup B)}.$$

We have $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = \frac{3}{4} + \frac{1}{3} - \frac{1}{4} = \frac{5}{6}$. So

$$P(A|C) = \frac{3/4}{5/6} = \frac{9}{10}.$$

We can also do this generically. We want to show

$$P(A) = P(A|B, C) < P(A|C) = \frac{P(A)}{P(A \cup B)}$$

so (rearranging the outer-most terms)

$$P(A) < \frac{P(A)}{P(A \cup B)} \qquad \text{which is just} \qquad P(A \cup B) < 1.$$

The last is true because $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A)P(B) = \frac{3}{4} + \frac{1}{3} - \frac{3}{4} \cdot \frac{1}{3} < 1$.

6. (10 points) Bob is playing a video game that has 7 levels. He starts at level 1, and has probability 0.2 of reaching level 2. In general, given that he reaches level j, he has probability 0.2 of reaching level $j+1$, for $1 \leq j \leq 6$. Let X be the highest level that he reaches. Find the PMF of X.

Solution:

Solution: This is similar to geometric random variable. Each time Bob makes it to a level, he gets stuck with probability 4/5 (or he finishes the game and is at level 7).

$$P(X=1) = P(\text{fails at level 1}) = 4/5,$$

$$P(X=2) = P(\text{pass at 1, fails at 2}) = P(\text{pass 1})P(\text{fails at 2}|\text{passes level 1}) = \frac{1}{5}\frac{4}{5} = \frac{4}{25}$$

$$P(X=3) = P(\text{pass 1, 2, fail at 3}) = P(\text{pass 1})P(\text{pass 2}|\text{pass 1})P(\text{fail 3}|\text{pass 2})$$
$$\frac{1}{5}\frac{1}{5}\frac{4}{5} = \frac{4}{5^3}.$$

This continues:

$$P(X=k) = \frac{4}{5^k} \qquad \text{for } k = 1,2,3,4,5,6$$

but

$$P(X=7) = \frac{1}{5^6}.$$

because you need to pass the first 6 levels.

SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM.
SCRATCH WORK WILL NOT BE GRADED

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Department of Mathematics, University of Wisconsin-Madison

Math 431 — Midterm Exam 1 — Spring 2025

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: Mikhail Feldman or Mikhail Ivanov

INSTRUCTIONS:

Time: **90 minutes**

- This exam contains 6 questions some with multiple parts, 10 pages (including the cover) for the total of 83 points. Read the problems carefully and budget your time wisely.
- You are allowed a single sheet of hand-written notes.
- **NO CALCULATORS** or other electronic devices are to be used. Turn off your phone so as to not disturb others.
- You **do not** need to simplify binomial coefficients or factorials, except when you are asked to do it.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded.
- Justify your steps.
- If you use an additional page for a particular problem, be sure to **CLEARLY** indicate this on the problem's page so I know to look further.

Question:	1	2	3	4	5	6	Total
Points:	16	12	8	18	21	8	83
Score:							

1. (16 points) The following questions are **Answers Only**.

   (a) In a five-card hand drawn at random from a well-shuffled standard deck, find the probability that the hand has exactly three aces.

   (b) Let X and Y be random variables taking only values 1 and 2. Give the precise conditions for X and Y to be independent.

   (c) We sample two numbers without replacement with order from the set $\{1, 2, 3, 4, 5, 6\}$. Describe the sample space of the experiment.

   (d) Suppose that A and B are events with $P(A) = \frac{1}{2}$, $P(B) = \frac{2}{3}$, and $P(A \cap B) = \frac{1}{4}$. Find the probability of $A^c \cup B$.

2. A certain candy company put golden tickets in the wrappers of 2% of their candy bars. If you find a golden ticket, you win a trip to their candy factory. Assume that every candy bar is equally likely to have a golden ticket.

   (a) (6 points) Suppose you bought 15 candy bars. What is the probability you find exactly 3 tickets?

   (b) (6 points) Suppose you buy candy bars until you find your first ticket. What is the probability that you must buy more than 50?

3. (8 points) A woman is pregnant with twin boys. Twins may be either identical or fraternal. Suppose that 1/3 of twins born are identical, that identical twins have a 50% chance of being both boys and a 50% chance of being both girls, and that for fraternal twins each twin independently has a 50% chance of being a boy and a 50% chance of being a girl. Given the above information, what is the probability that the woman's twins are identical?

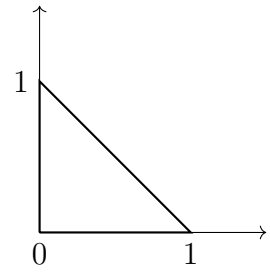
4. A box contains 8 green balls, 6 red balls, and 4 yellow balls. Two balls are sampled uniformly at random without replacement.

   (a) (6 points) What is the probability that one ball is green and another is yellow? (First write down your sample space.)

   (b) (6 points) What is the probability that at least one of the balls is red?

(c) (6 points) Let X be the number of red balls in the sample. Find CDF of X and draw the graph of this function.

5. We choose a point (X, Y) uniformly from the triangle with vertices $(0, 0)$, $(0, 1)$ and $(1, 0)$. Let $Z = 2X - 1$.



(a) (4 points) Find the range of the random variable Z.

(b) (5 points) Compute $P(Z > \frac{2}{3})$.

(c) (6 points) Compute the cumulative distribution function of Z.

(d) (6 points) Find the probability density function of Z.

6. (8 points) Professor May B. Right often has her facts wrong, and answers each of her students' questions incorrectly with probability 1/4, independent of other questions. In each lecture, May is asked 0, 1, or 2 questions with equal probability 1/3. Find the probability that she gave at least one wrong answer during one lecture.

SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM.
SCRATCH WORK WILL NOT BE GRADED

Department of Mathematics, University of Wisconsin-Madison

Math 431 — Midterm Exam 1 — Solutions — Spring 2025

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: Mikhail Feldman or Mikhail Ivanov

INSTRUCTIONS:

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- This exam contains 6 questions some with multiple parts, 10 pages (including the cover) for the total of 83 points. Read the problems carefully and budget your time wisely.
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Question:	1	2	3	4	5	6	Total
Points:	16	12	8	18	21	8	83
Score:							

1. (16 points) The following questions are **Answers Only**.

(a) In a five-card hand drawn at random from a well-shuffled standard deck, find the probability that the hand has exactly three aces.

Solution:

$$\frac{\binom{4}{3}\binom{48}{2}}{\binom{52}{5}}.$$

(b) Let X and Y be random variables taking only values 1 and 2. Give the precise conditions for X and Y to be independent.

Solution:

$$\begin{aligned}
P(X=2,Y=2) &= P(X=2)P(Y=2)\\
P(X=1,Y=2) &= P(X=1)P(Y=2)\\
P(X=2,Y=1) &= P(X=2)P(Y=1)\\
P(X=1,Y=1) &= P(X=1)P(Y=1)
\end{aligned}$$

One of equalities is enough.

(c) We sample two numbers without replacement with order from the set $\{1,2,3,4,5,6\}$. Describe the sample space of the experiment.

Solution: $\Omega = \{(s_1,s_2):\ s_1,s_2 \in \{1,2,3,4,5,6\},\ \ s_1 \neq s_2\}$

(d) Suppose that A and B are events with $P(A)=\frac{1}{2}$, $P(B)=\frac{2}{3}$, and $P(A\cap B)=\frac{1}{4}$. Find the probability of $A^c \cup B$.

Solution: We will use the formula:

$$P(A^c \cup B) = P(A^c) + P(B) - P(A^c \cap B).$$

First we find $P(A^c) = 1 - P(A) = \frac{1}{2}$.
Next we find $P(A^c \cap B)$. Since $B = (A\cap B)\cup(A^c\cap B)$ and the sets $A\cap B$ and $A^c\cap B$ are disjoint, then $P(A^c\cap B) = P(B) - P(A\cap B) = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$.
Now we have

$$P(A^c \cup B) = P(A^c) + P(B) - P(A^c \cap B) = \frac{1}{2} + \frac{2}{3} - \frac{5}{12} = \frac{3}{4}.$$

2. A certain candy company put golden tickets in the wrappers of 2% of their candy bars. If you find a golden ticket, you win a trip to their candy factory. Assume that every candy bar is equally likely to have a golden ticket.

   (a) (6 points) Suppose you bought 15 candy bars. What is the probability you find exactly 3 tickets?

Solution: Let the random variable X be the number of tickets in 15 candy bars. Then X counts the number of successes in 15 independent trials with success probability $p = \frac{2}{100} = \frac{1}{50}$. Then $X \sim \text{Bin}(15, \frac{1}{50})$, and the probability you find exactly 3 tickets is

$$P(X = 3) = \binom{15}{3} \left(\frac{1}{50}\right)^3 \left(\frac{49}{50}\right)^{12}.$$

   (b) (6 points) Suppose you buy candy bars until you find your first ticket. What is the probability that you must buy more than 50?

Solution: You must buy more than 50 candy bars to find your first ticket only in the case when there is no tickets in the first 50 candy bars, which has the probability

$$(1-p)^{50} = \left(\frac{49}{50}\right)^{50}.$$

3. (8 points) A woman is pregnant with twin boys. Twins may be either identical or fraternal. Suppose that 1/3 of twins born are identical, that identical twins have a 50% chance of being both boys and a 50% chance of being both girls, and that for fraternal twins each twin independently has a 50% chance of being a boy and a 50% chance of being a girl. Given the above information, what is the probability that the woman's twins are identical?

Solution: By Bayes' rule,

$$P(\text{identical}|\text{B and B}) = \frac{P(\text{B and B}|\text{identical})P(\text{identical})}{P(\text{B and B})} = \frac{\frac{1}{2}\cdot\frac{1}{3}}{\frac{1}{2}\cdot\frac{1}{3}+\frac{1}{4}\cdot\frac{2}{3}} = \frac{1}{2}.$$

4. A box contains 8 green balls, 6 red balls, and 4 yellow balls. Two balls are sampled uniformly at random without replacement.

   (a) (6 points) What is the probability that one ball is green and another is yellow? (First write down your sample space.)

Solution: The total number of balls is $8+6+4=18$. The number of ways to choose two balls out of 18 without replacement is $\binom{18}{2}$. Similarly, the number of ways to choose 1 green ball out of 8 and one yellow ball out of 4 are $\binom{8}{1}$ and $\binom{4}{1}$ respectively. Then

$$P(1 \text{ green, } 1 \text{ yellow}) = \frac{\binom{8}{1}\binom{4}{1}}{\binom{18}{2}} = \frac{32}{153}.$$

   (b) (6 points) What is the probability that at least one of the balls is red?

Solution: The number of pairs in which there are no red balls is $\binom{12}{2}$, where 12=8+4 is the total number of green and yellow balls. Then

$$P(\text{at least 1 red ball}) = 1 - P(\text{no red balls}) = 1 - \frac{\binom{12}{2}}{\binom{18}{2}} = \frac{87}{153} = \frac{29}{51}.$$

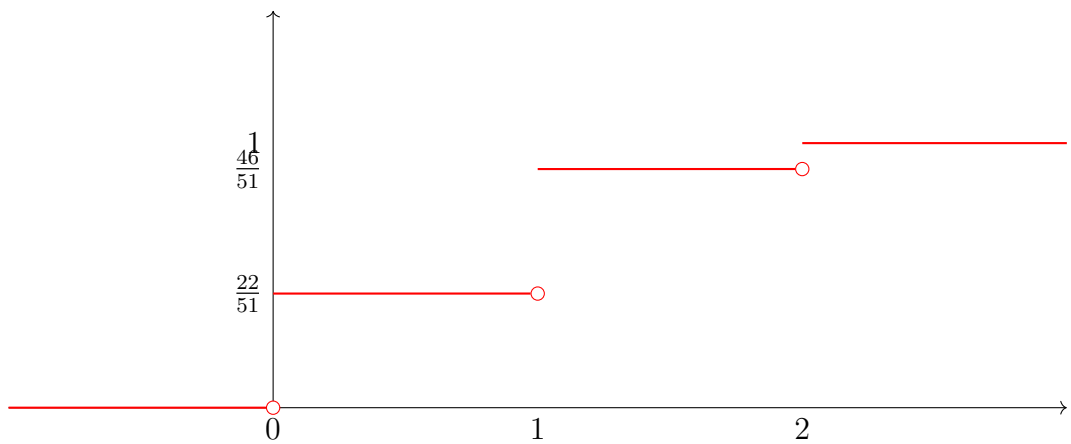
(c) (6 points) Let X be the number of red balls in the sample. Find CDF of X and draw the graph of this function.

Solution: X has the hypergeometric distribution with parameters $(18, 6, 2)$. Then the range of X is $\{0, 1, 2\}$, and

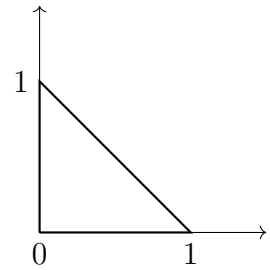
$$P(X=0) = \frac{\binom{6}{0}\binom{12}{2}}{\binom{18}{2}} = \frac{22}{51}; \quad P(X=1) = \frac{\binom{6}{1}\binom{12}{1}}{\binom{18}{2}} = \frac{8}{17}; \quad P(X=2) = \frac{\binom{6}{2}\binom{12}{0}}{\binom{18}{2}} = \frac{5}{51}.$$

Then CDF of X, the function $F(s) = P(X \le s)$, is

$$F(s) = \begin{cases} 0, & s < 0; \\ \frac{22}{51}, & 0 \le s < 1; \\ \frac{46}{51}, & 1 \le s < 2; \\ 1, & s \ge 2. \end{cases}$$



5. We choose a point (X, Y) uniformly from the triangle with vertices $(0,0)$, $(0,1)$ and $(1,0)$. Let $Z = 2X - 1$.

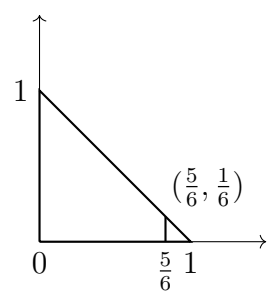


(a) (4 points) Find the range of the random variable Z.

Solution: Range of X is $[0,1]$, and so the range of $Z = 2X - 1$ is $[-1,1]$.

(b) (5 points) Compute $P(Z > \frac{2}{3})$.

Solution:



Denote by A the triangle defined in the problem. The region where $Z > \frac{2}{3}$ corresponds to the region where $2X - 1 > \frac{2}{3}$, which simplifies to $X > \frac{5}{6}$ within triangle A. Thus, the region $\{X > \frac{5}{6}\}$ forms triangle B with vertices $\left(\frac{5}{6}, 0\right)$, $(1,0)$, and $\left(\frac{5}{6}, \frac{1}{6}\right)$. This is a right triangle with two equal sides of length $\frac{1}{6}$. Then we have:

$$P\left(Z > \frac{2}{3}\right) = \frac{\text{area of triangle } B}{\text{area of triangle } A} = \frac{\frac{1}{2} \cdot \frac{1}{6} \cdot \frac{1}{6}}{\frac{1}{2} \cdot 1 \cdot 1} = \frac{1}{36}.$$

(c) (6 points) Compute the cumulative distribution function of Z.

Solution: Since the range of Z is $[-1, 1]$, then cumulative distribution function $F(s)$ of Z satisfies $F(s) = 0$ for $s < -1$ and $F(s) = 1$ for $s \geq 1$.

Let $s \in [-1, 1)$. $F(s) = P(Z \leq s)$. We will first find $P(Z > s)$.

Similar to part (b) of this problem, the region where $Z > s$ corresponds to the region where $2X - 1 > s$, which simplifies to $X > \frac{s+1}{2}$ within triangle A. Note that $\frac{s+1}{2} \in [0, 1)$ since $s \in [-1, 1)$.

Thus, the region $\{Z > s\}$ forms triangle B_s with vertices $\left(\frac{s+1}{2}, 0\right)$, $(1, 0)$, and $\left(\frac{s+1}{2}, \frac{1-s}{2}\right)$, where $\frac{1-s}{2} = 1 - \frac{s+1}{2}$. This is a right triangle with two equal sides of length $\frac{1-s}{2}$. Then we have:

$$P(Z > s) = \frac{\text{area of triangle } B_s}{\text{area of triangle } A} = \frac{\frac{1}{2} \cdot \frac{1-s}{2} \cdot \frac{1-s}{2}}{\frac{1}{2} \cdot 1 \cdot 1} = \frac{(1-s)^2}{4}.$$

From this

$$F(s) = P(Z \leq s) = 1 - P(Z > s) = 1 - \frac{(1-s)^2}{4}.$$

Then the cumulative distribution function of Z is:

$$F(s) = \begin{cases} 0, & s < -1; \\ 1 - \frac{(1-s)^2}{4}, & -1 \leq s < 1; \\ 1, & s \geq 1. \end{cases}$$

(d) (6 points) Find the probability density function of Z.

Solution: Cumulative distribution function $F(s)$ is differentiable everywhere except $s = -1$ and $s = 1$. Then the probability density function $f(s)$ of Z is $f(s) = F'(s)$ at every point except $s = -1$ and $s = 1$, and defined arbitrarily at these two points, so

$$f(s) = \begin{cases} 0, & s < -1; \\ \frac{1-s}{2}, & -1 \leq s < 1; \\ 0, & s \geq 1. \end{cases}$$

6. (8 points) Professor May B. Right often has her facts wrong, and answers each of her students' questions incorrectly with probability 1/4, independent of other questions. In each lecture, May is asked 0, 1, or 2 questions with equal probability 1/3. Find the probability that she gave at least one wrong answer during one lecture.

Solution: Let N, be the number of questions that professor was asked and A be the event that she gave at least one wrong answer during one lecture. We know that $\{N = 0\}$, $\{N = 1\}$, $\{N = 2\}$ is a partition, so

$$P(A) = P(N = 0)P(A|N = 0) + P(N = 1)P(A|N = 1) + P(N = 2)P(A|N = 2)$$

We know that $P(N = 0) = P(N = 1) = P(N = 2) = \frac{1}{3}$, $P(A|N = 0) = 0$ (there is no questions to answer), $P(A|N = 1) = \frac{1}{4}$ and

$$P(A|N = 2) = 1 - P(\text{both questions were answered correctly}) = 1 - \left(\frac{3}{4}\right)^2,$$

so, combining together

$$P(A) = \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{7}{16} = \frac{11}{48}.$$

OR

We can compute $P(A^c)$, i.e the probability that all answers were correct:

$$\begin{aligned} P(A^c) &= P(N = 0)P(A^c|N = 0) + P(N = 1)P(A^c|N = 1) + P(N = 2)P(A^c|N = 2) \\ &= \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{3} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{16 + 12 + 9}{48} = \frac{37}{48}. \end{aligned}$$

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